Fuzzy Inference Systems

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Lecture outline

Fuzzy Rules

Fuzzy Inference

- Mamdani-Style Inference
- Takagi-Sugeno-Kang-Style Inference

Building A Fuzzy Expert System

What is a fuzzy rule?

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A fuzzy rule can be defined as a conditional statement in the form

$$\begin{array}{ll} \text{IF} & x \text{ is } A \\ \text{THEN} & y \text{ is } B \end{array}$$

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A fuzzy rule can be defined as a conditional statement in the form

$$\begin{array}{ll} \text{IF} & x \text{ is } A \\ \text{THEN} & y \text{ is } B \end{array}$$

x, y : linguistic variables

A, B : linguistic values

Classical rules

A classical rule uses binary logic.

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Rule 1

IF	speed is > 100
THEN	stopping_distance is long

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Rule 2

IF speed is < 400

THEN stopping_distance is short

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- The variable 'speed' can have any numerical value.

Classical rules

A classical rule uses binary logic.

Rule 1

IF speed is > 100

THEN stopping_distance is long

Rule 2

IF speed is < 400

THEN stopping_distance is short

- The variable 'speed' can have any numerical value.
- But the linguistic variable 'stopping_distance' can only take the value of either 'long' or 'short'.

Fuzzy rules

Using the same variables,

Fuzzy rules

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Rule 1

IF	speed is fast
THEN	stopping_distance is long

Fuzzy rules

Using the same variables,

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IF speed is fast

THEN stopping_distance is long

Rule 2

IF speed is slow

THEN stopping_distance is short

Fuzzy rules

Using the same variables,

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- The linguistic variable 'speed' can also take any numerical value (translated to the corresponding fuzzy sets: 'slow', 'medium', and 'fast').

Fuzzy rules

Using the same variables,

Rule 1

IF speed is fast

THEN stopping_distance is long

Rule 2

IF speed is slow

THEN stopping_distance is short

- The linguistic variable 'speed' can also take any numerical value (translated to the corresponding fuzzy sets: 'slow', 'medium', and 'fast').
- The linguistic variable 'stopping_distance' can be numerical value (translated from the corresponding fuzzy sets: 'short', 'medium', and 'long').

Fuzzy reasoning

Classical rule-based systems:

Rule 1

IF	speed is $>$ 100
THEN	stopping_distance is long

Fuzzy reasoning

Classical rule-based systems:

Rule 1

IF	speed is > 100	$\rightarrow$ antecedent
THEN	stopping_distance is long	

Fuzzy reasoning

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Fuzzy reasoning

Classical rule-based systems:

Rule 1

IF	speed is > 100	$\rightarrow$ antecedent
THEN	stopping_distance is long	$\rightarrow$ consequent

If the rule antecedent is **true**, then the consequent is **true**.

Fuzzy systems:

Rule 1	
IF	speed is fast
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Fuzzy reasoning

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Fuzzy reasoning

Fuzzy systems:

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IF	speed is fast	$\rightarrow$ antecedent
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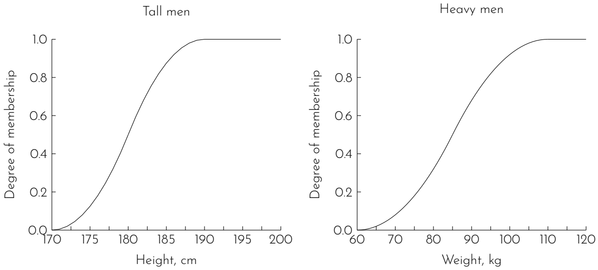
If the rule antecedent is **true to some degree of membership**, then the consequent is **true to the same degree**.

Example 1

Consider two fuzzy sets 'tall men' and 'heavy men'.



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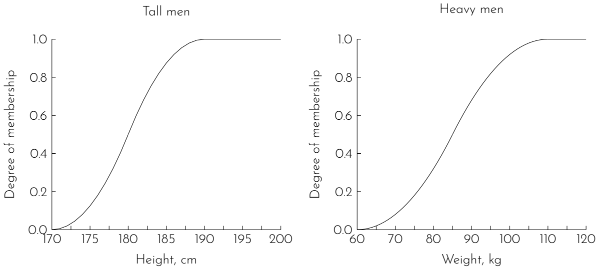


Given the relationship between a man's height and his weight,

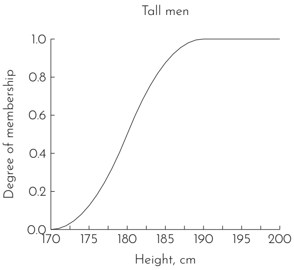
Rule 1

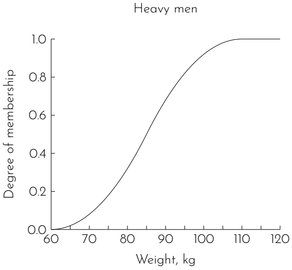
IF height is tall

THEN weight is heavy

Example 1

For a man with height 181 cm, his weight can be derived from the fuzzy sets.





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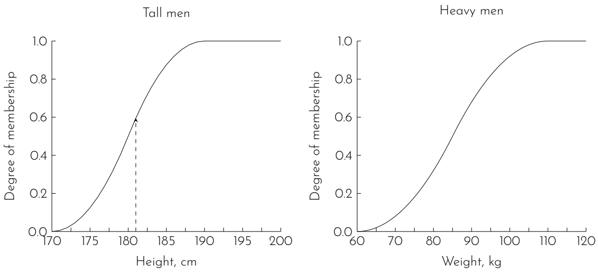

$$181 \text{ cm} \rightarrow 0.595$$

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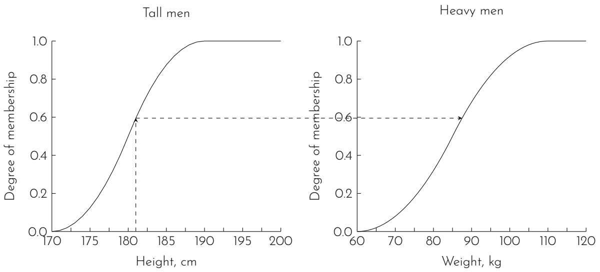

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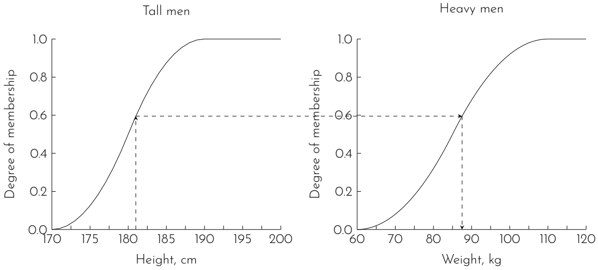

$$181 \text{ cm} \to 0.595 \to 87.5 \text{ kg}$$

Fuzzy rules

Fuzzy rule with multiple parts of antecedent:

Fuzzy rules

Fuzzy rule with multiple parts of antecedent:

IF	project_duration is long
AND	project_staffing is large
AND	project_funding is inadequate
THEN	risk is high

Fuzzy rules

Fuzzy rule with multiple parts of antecedent:

IF	project_duration is long		IF	service is excellent
AND	project_staffing is large		OR	food is delicious
AND	project_funding is inadequate		THEN	tip is generous
THEN	risk is high			

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All parts of the antecedent are calculated and resolved to a single number using fuzzy set operations.

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All parts of the antecedent are calculated and resolved to a single number using fuzzy set operations.

$$A \text{ AND } B = \min(\mu_A, \mu_B)$$

$$A \text{ OR } B = \max(\mu_A, \mu_B)$$

Fuzzy rules

Fuzzy rule with multiple parts of consequent:

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Fuzzy rule with multiple parts of consequent:

IF	temperature is hot
THEN	hot_water is reduced;
	cold_water is increased

Fuzzy rules

Fuzzy rule with multiple parts of consequent:

IF	temperature is hot
THEN	hot_water is reduced;
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All parts of the consequent are affected equally by the antecedent.

Fuzzy inference

Fuzzy inference is a process of mapping from a given input to an output, using the theory of fuzzy set.

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Step 4: Defuzzification

Fuzzy inference

There are two different techniques for fuzzy inference.

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Mamdani-style inference

Takagi-Sugeno-Kang-style inference

Fuzzy inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

The rule consequent is a fuzzy set.

Takagi-Sugeno-Kang-style inference

Fuzzy inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

The rule consequent is a fuzzy set.

$$\begin{array}{rl} \text{IF} & X_1 \text{ is } A_1 \text{ and } \ldots \text{and } X_n \text{ is } A_n \\ \text{THEN} & Y \text{ is } B_i \end{array}$$

Takagi-Sugeno-Kang-style inference

Fuzzy inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

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Takagi-Sugeno-Kang-style inference

The rule consequent is a mathematical function.

Fuzzy inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

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Takagi-Sugeno-Kang-style inference

The rule consequent is a mathematical function.

$$\begin{aligned} \text{IF} \quad & X_1 \text{ is } A_1 \text{ and } \ldots \text{and } X_n \text{ is } A_n \\ \text{THEN} \quad & Y = p_0 + p_1 X_1 + \ldots + p_n X_n \end{aligned}$$

Fuzzy inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

The rule consequent is a fuzzy set.

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The rule consequent is a mathematical function.

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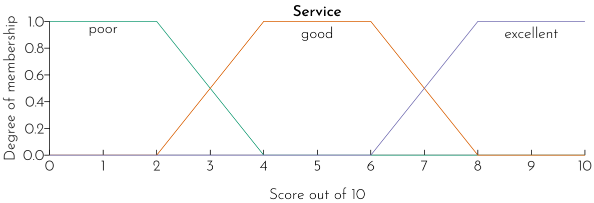
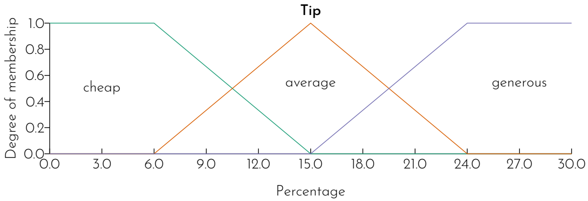
Mamdani-style is the more commonly used method.

Mamdani-style inference

Given a fuzzy system with two linguistic variables 'service' and 'tip' with the following fuzzy sets.





Mamdani-style inference

The following Mamdani-style rule applies to the system.

IF	service is good
THEN	tip is average

If a customer then rates the service at the restaurant as being a 3 out of 10, what would be the percentage of the given tip?

Mamdani-style inference

Step 1: Fuzzification

Mamdani-style inference

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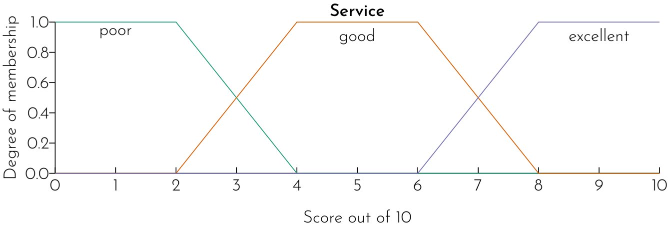

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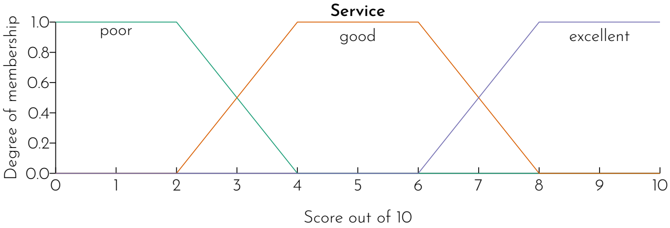

Let X denotes the linguistic variable 'service'.

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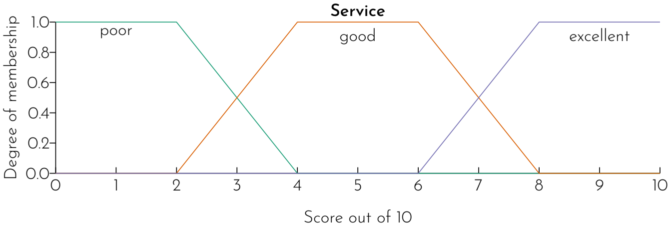

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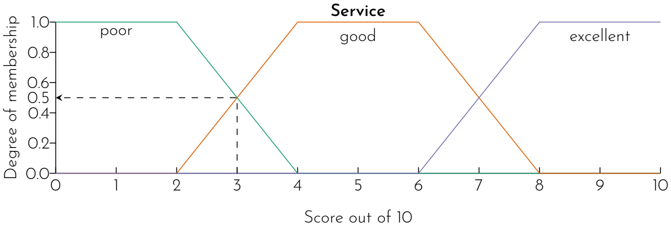

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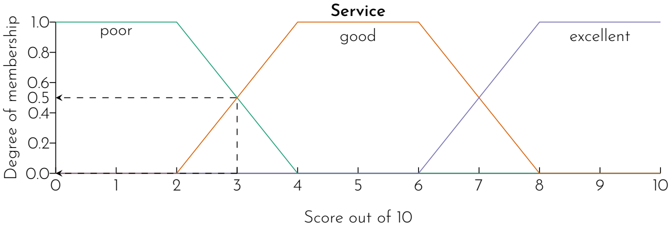

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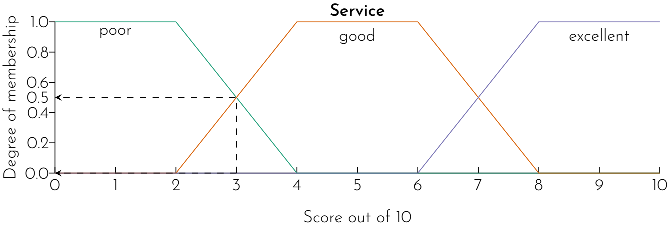

Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

$$\mu_{X=\text{poor}} = 0.5$$
$$\mu_{X=\text{good}} = 0.5$$
$$\mu_{X=\text{excellence}} = 0.0$$

Mamdani-style inference

Step 2: Rule evaluation

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Fire the rule to the same degree.

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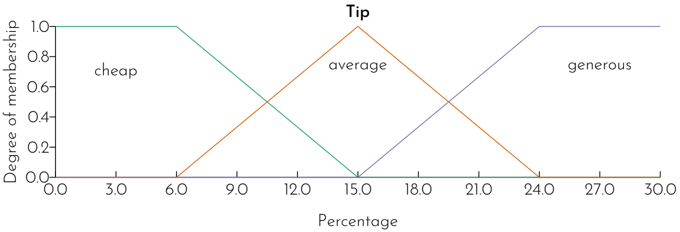

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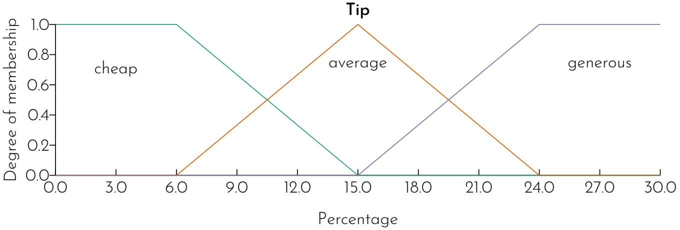

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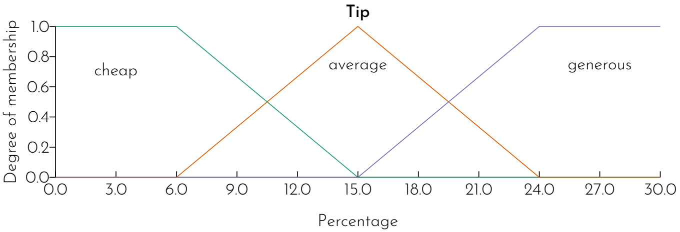

IF service is good $\rightarrow \mu_{X=\text{good}} = 0.5$

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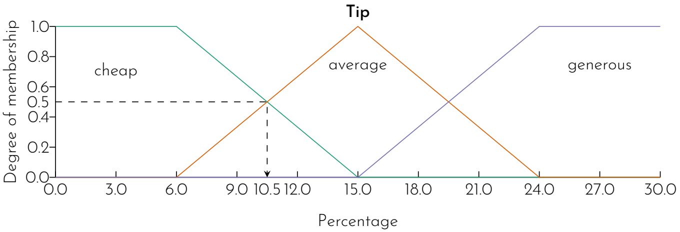

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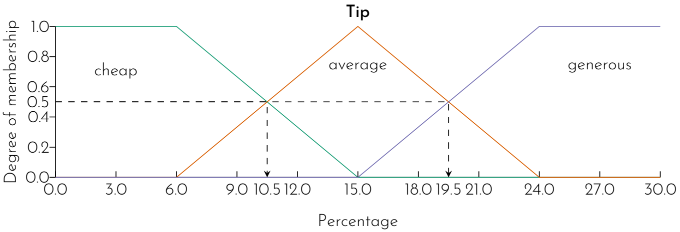

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Mamdani-style inference

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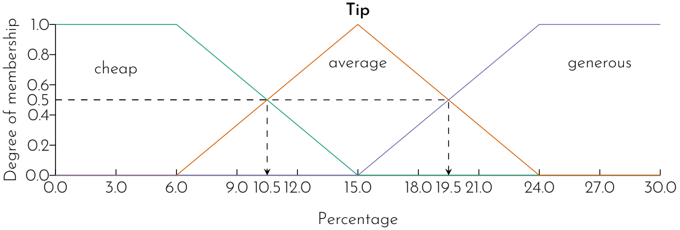

IF	service is good	$\rightarrow \mu_{X=\text{good}} = 0.5$
THEN	tip is average	

The tip amount in 'average' is between 10.5% and 19.5%.

Mamdani-style inference

Step 2: Rule evaluation (continued)

Mamdani-style inference

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The membership function of the consequent is then 'clipped' or 'scaled'.

Mamdani-style inference

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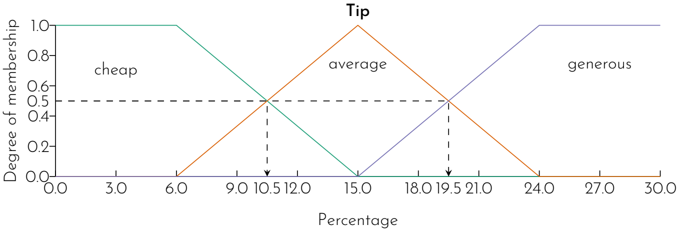

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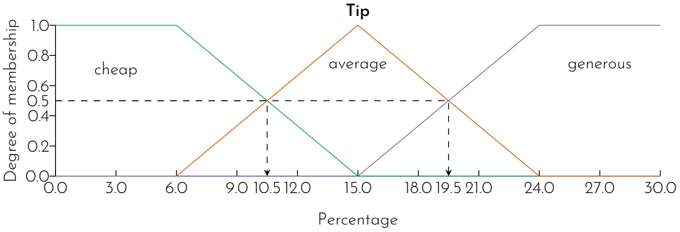

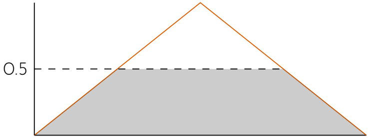

Clipped

Mamdani-style inference

Step 2: Rule evaluation (continued)

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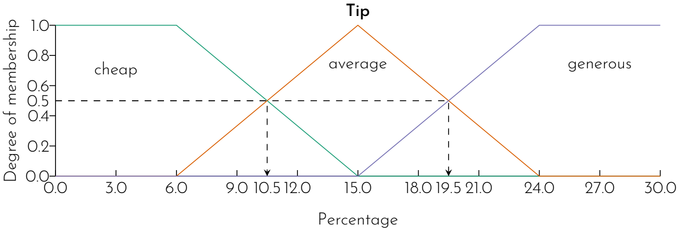

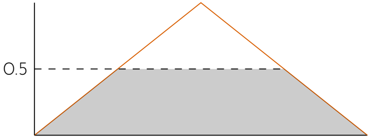

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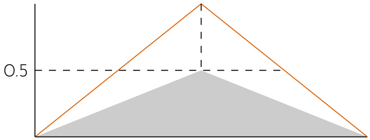

Scaled

Mamdani-style inference

Step 3: Aggregation of rule consequents

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Mamdani-style inference

Step 3: Aggregation of rule consequents

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- This step is important if multiple rules are fired.
- We will look at this in the latter example.

Step 4: Defuzzification

Mamdani-style inference

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- Identify the crisp number to be the final output of the fuzzy system.

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Mamdani-style inference

Step 4: Defuzzification

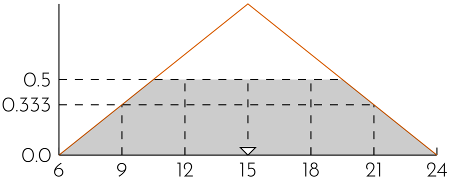
- Identify the crisp number to be the final output of the fuzzy system.
- The centroid technique is commonly used to identify the crisp output.
- The centroid (centre of gravity) is identified using

$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A}$$

Step 4: Defuzzification (continued)

Mamdani-style inference

Step 4: Defuzzification (continued)



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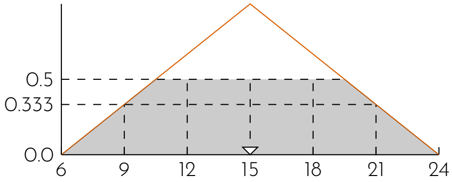
Mamdani-style inference

Step 4: Defuzzification (continued)



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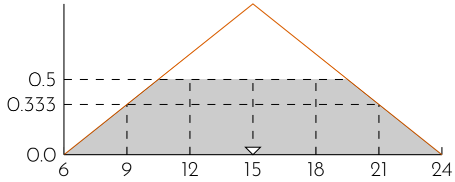
$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A}$$

Mamdani-style inference

Step 4: Defuzzification (continued)



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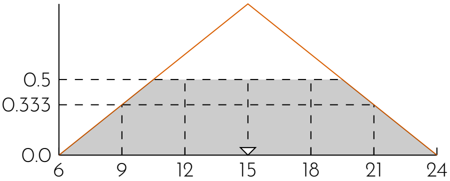
$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(6+24)(0)+(9+21)(0.333)+(12+15+18)(0.5)}{2(0)+2(0.333)+3(0.5)}$$

Mamdani-style inference

Step 4: Defuzzification (continued)



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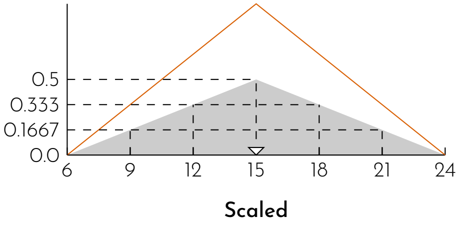
$$= 15$$

Mamdani-style inference

Step 4: Defuzzification (continued)

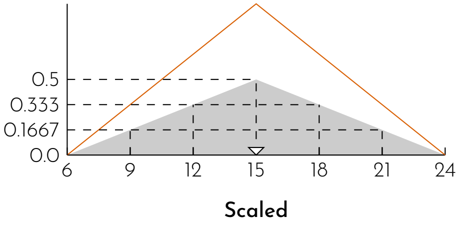
Mamdani-style inference

Step 4: Defuzzification (continued)



Mamdani-style inference

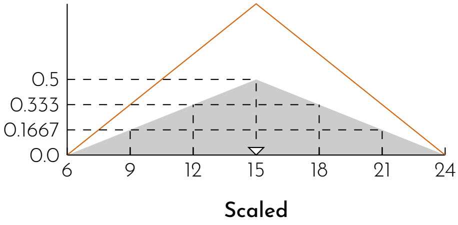
Step 4: Defuzzification (continued)



$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A}$$

Mamdani-style inference

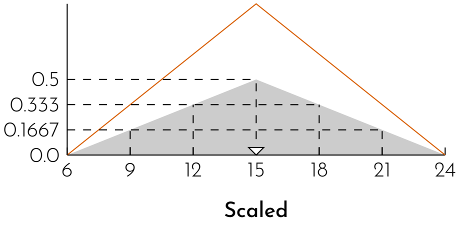
Step 4: Defuzzification (continued)



$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(6+24)(0)+(9+21)(0.1667)+(12+18)(0.333)+15(0.5)}{2(0)+2(0.1667)+2(0.333)+0.5}$$

Mamdani-style inference

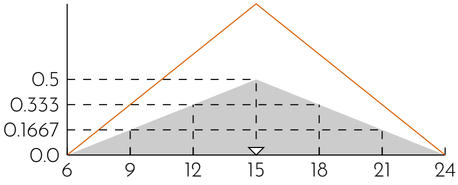
Step 4: Defuzzification (continued)



$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(6+24)(0)+(9+21)(0.1667)+(12+18)(0.333)+15(0.5)}{2(0)+2(0.1667)+2(0.333)+0.5}$$

$$= 15$$

Step 4: Defuzzification (continued)



$$\text{centre of gravity} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(6+24)(0)+(9+21)(0.1667)+(12+18)(0.333)+15(0.5)}{2(0)+2(0.1667)+2(0.333)+0.5}$$

$$= 15$$

The final output of the fuzzy system, i.e. the percentage of the tip that would be given is 15%.

Takagi-Sugeno-Kang-style inference

Sugeno-style inference shortens the time of fuzzy inference by using a singleton as the membership function of the rule consequent.

Takagi-Sugeno-Kang-style inference

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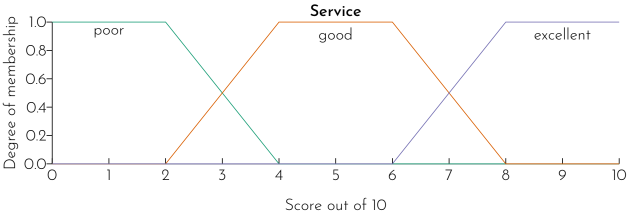
The previous example is modified slightly for Sugeno-style inference.

Sugeno-style inference

We have the linguistic variable ‘service’ with the same fuzzy sets

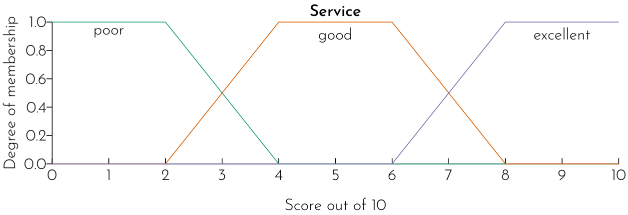
Sugeno-style inference

We have the linguistic variable 'service' with the same fuzzy sets



Sugeno-style inference

We have the linguistic variable 'service' with the same fuzzy sets

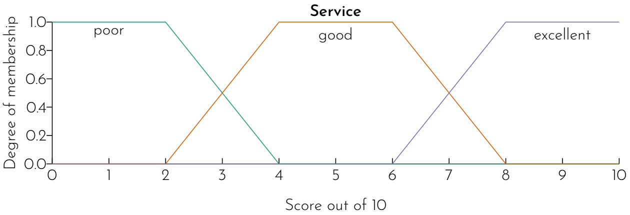


with the Sugeno-style rule

IF	service is good
THEN	tip is $3\times$service %

Sugeno-style inference

We have the linguistic variable 'service' with the same fuzzy sets



with the Sugeno-style rule

IF	service is good
THEN	tip is $3 \times$service %

If a customer then rates the service at the restaurant as being a 3 out of 10, what would be the percentage of the given tip?

Sugeno-style inference

Step 1: Fuzzification

Sugeno-style inference

Step 1: Fuzzification

Identify the degree of membership of the input variable in each fuzzy set.

Sugeno-style inference

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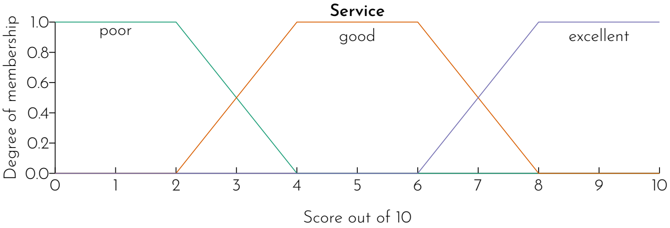

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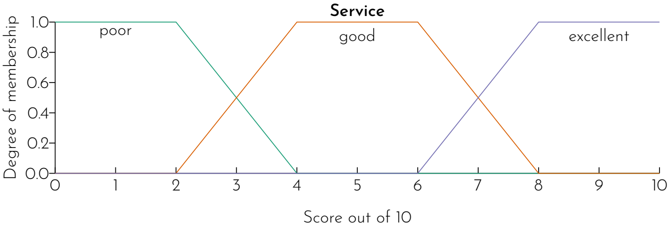

Let X denotes the linguistic variable 'service'.

Sugeno-style inference

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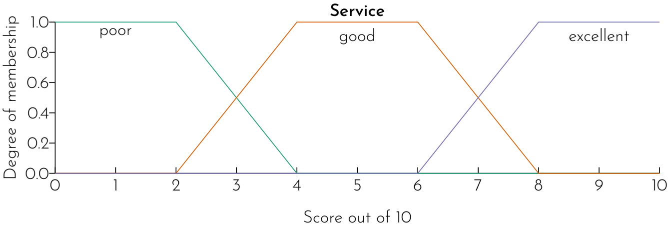

Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

Sugeno-style inference

Step 1: Fuzzification

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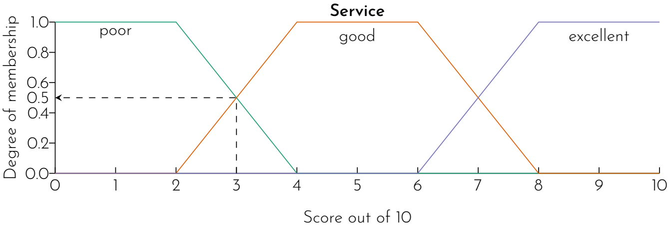

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Step 1: Fuzzification

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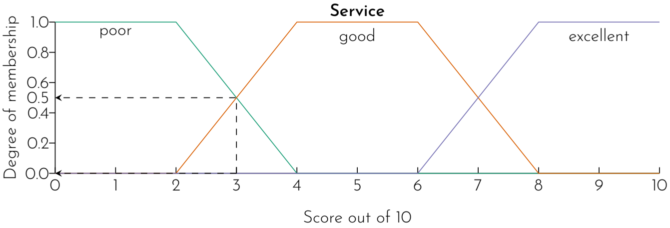

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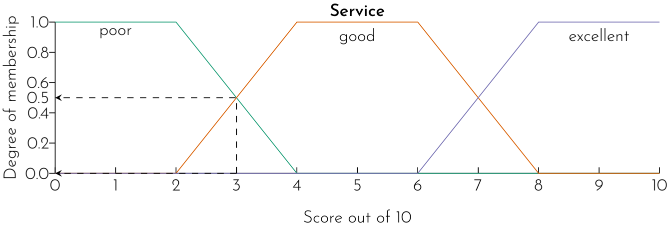
Step 1: Fuzzification

Identify the degree of membership of the input variable in each fuzzy set.



Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

$$\mu_{X=\text{poor}} = 0.5$$
$$\mu_{X=\text{good}} = 0.5$$
$$\mu_{X=\text{excellence}} = 0.0$$

Sugeno-style inference

Step 2: Rule evaluation

Sugeno-style inference

Step 2: Rule evaluation

According to this rule:

IF	service is good	$\rightarrow \mu_{X=\text{good}} = 0.5$
THEN	tip is $3\times$service %	

Sugeno-style inference

Step 2: Rule evaluation

According to this rule:

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tip $= 3 \times 3 = 9\%$

Sugeno-style inference

Step 2: Rule evaluation

According to this rule:

IF	service is good	$\rightarrow \mu_{X=\text{good}} = 0.5$
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tip $= 3 \times 3 = 9\%$

the firing strength of this rule is $\alpha = \mu_{X=\text{good}} = 0.5$

Sugeno-style inference

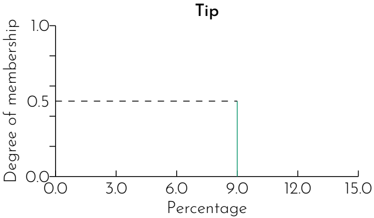
Step 2: Rule evaluation

According to this rule:

$$\begin{array}{lll} \text{IF} & \text{service is good} & \rightarrow \mu_{X=\text{good}} = 0.5 \\ \text{THEN} & \text{tip is } 3\times\text{service } \% & \end{array}$$

tip $= 3 \times 3 = 9\%$

the firing strength of this rule is $\alpha = \mu_{X=\text{good}} = 0.5$



Sugeno-style inference

Step 3: Aggregation of rule consequents

- Skip the step as there is only one rule fired.

Step 4: Defuzzification

Sugeno-style inference

Step 4: Defuzzification

The final crisp output is obtained using the weighted average formula.

Sugeno-style inference

Step 4: Defuzzification

The final crisp output is obtained using the weighted average formula.

$$\text{weighted average} = \frac{\displaystyle\sum_{x=a}^{b} \alpha_x \cdot x}{\displaystyle\sum_{x=a}^{b} \alpha_x}$$

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Step 4: Defuzzification

The final crisp output is obtained using the weighted average formula.

$$\text{weighted average} = \frac{\sum_{x=a}^{b} \alpha_x \cdot x}{\sum_{x=a}^{b} \alpha_x}$$

$$\therefore \text{tip} = \frac{0.5 \cdot 9}{0.5} = 9\%$$

Example 2

Consider the fuzzy system with two input variables x and y, and one output variable z. All the variables have the range between 0 and 10.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$
z	C_1	$(1/2, 0/5)$
z	C_2	$(0/2, 1/4, 1/6, 0/8)$
z	C_3	$(0/7, 1/8)$

Example 2

The following Mamdani-style rules apply to this system.

Rule 1

IF	x is A_3
OR	y is B_1
THEN	z is C_1

Rule 2

IF	x is A_2
AND	y is B_2
THEN	z is C_2

Rule 3

IF	x is A_1
THEN	z is C_3

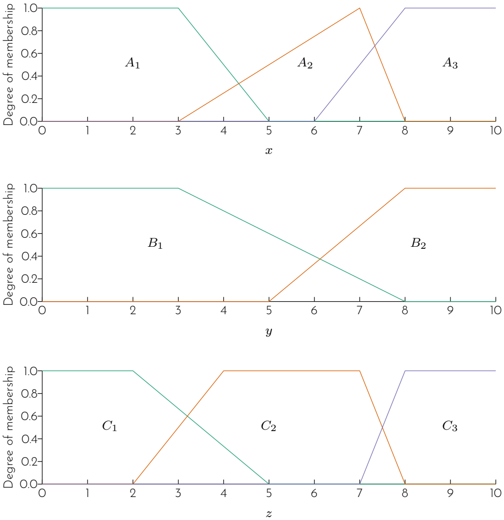
Given that x_1 is 3.5, y_1 is 6.5, what is the value of z_1?

Example 2

First, draw the membership function of the fuzzy sets on the graph paper.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$
z	C_1	$(1/2, 0/5)$
z	C_2	$(0/2, 1/4, 1/6, 0/8)$
z	C_3	$(0/7, 1/8)$

Example 2

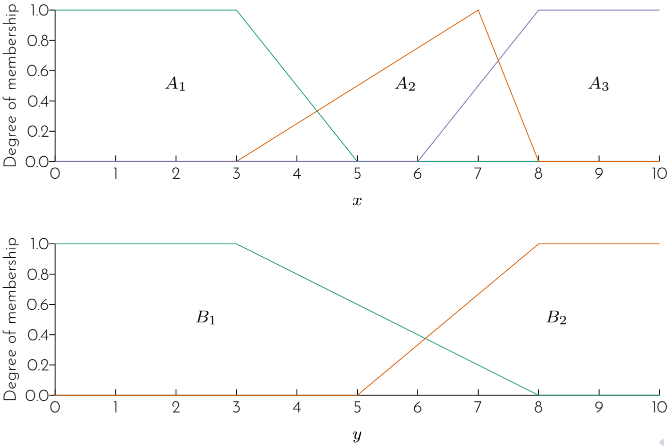


Degree of membership
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
6
7
8
9
10
x
A_1
A_2
A_3
Degree of membership
y
B_1
B_2
Degree of membership
z
C_1
C_2
C_3

Example 2

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

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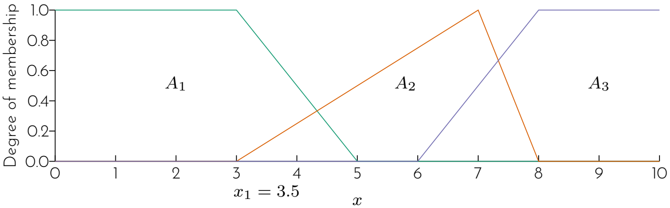

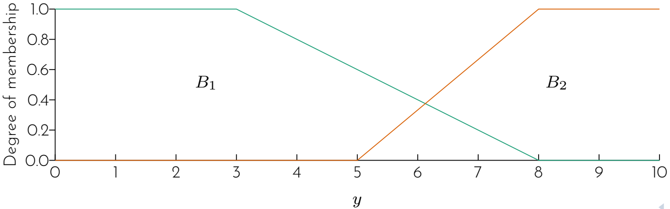

Example 2

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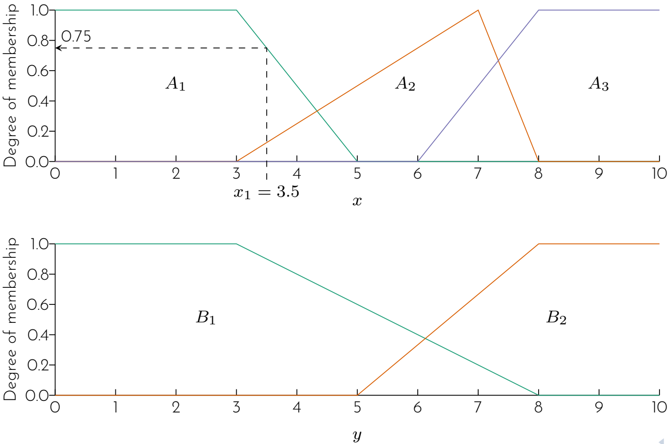

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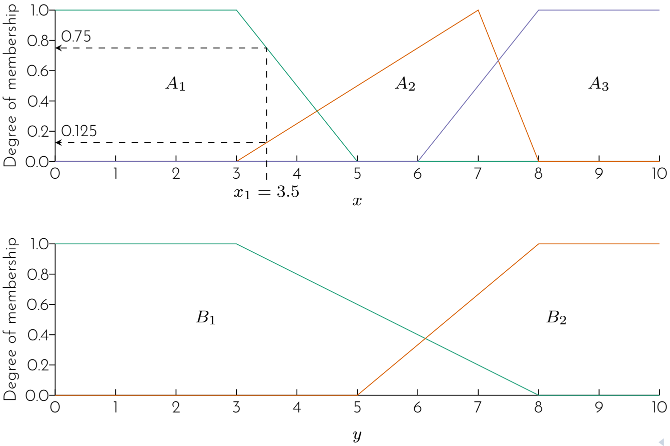

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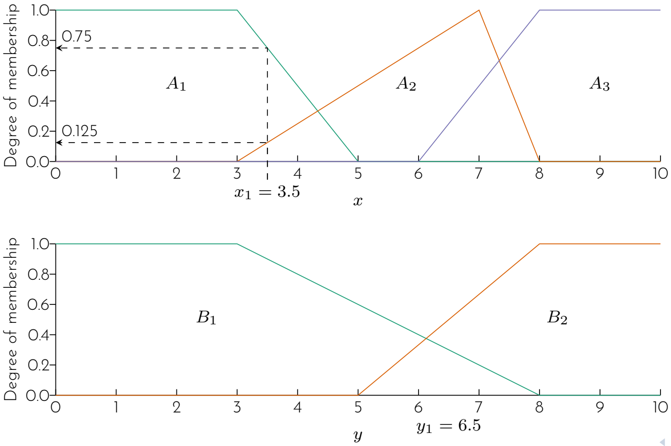

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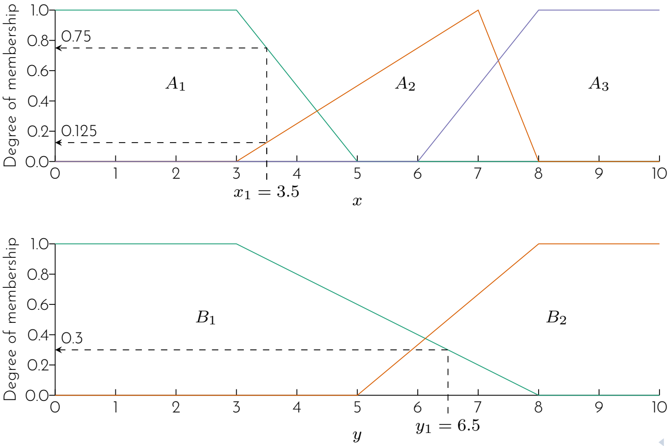

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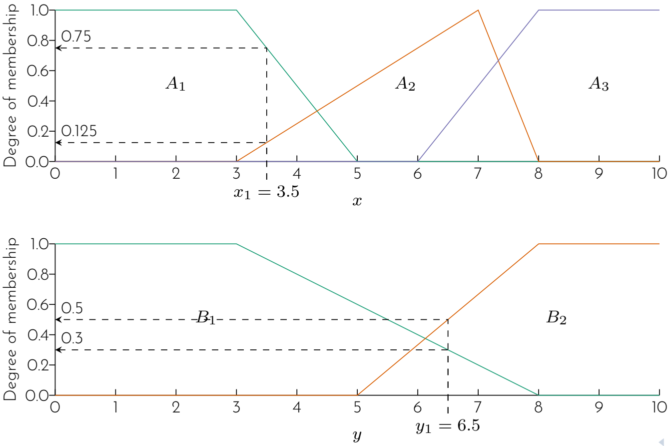


Example 2

Step 1: Fuzzification (continued)

Therefore we have

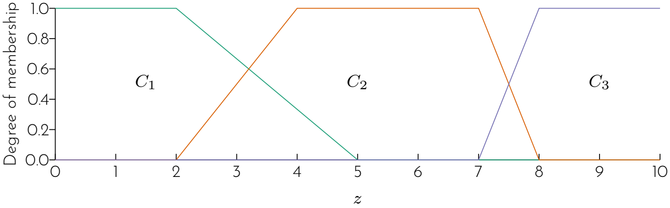
$$\mu_{x=A_1} = 0.75$$
$$\mu_{x=A_2} = 0.125$$
$$\mu_{x=A_3} = 0$$

$$\mu_{y=B_1} = 0.3$$
$$\mu_{y=B_2} = 0.5$$

Example 2

Step 2: Rule evaluation

Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	



Example 2

Step 2: Rule evaluation

Rule 1		Degree of membership
IF	x is A_3	0.0
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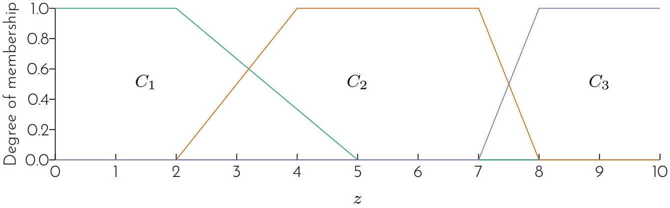
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



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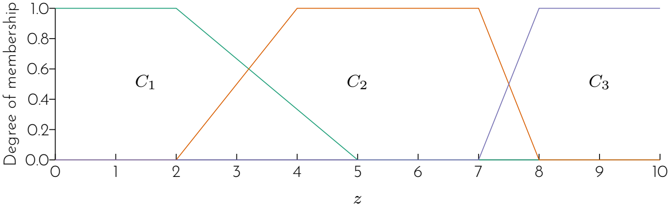
0.3



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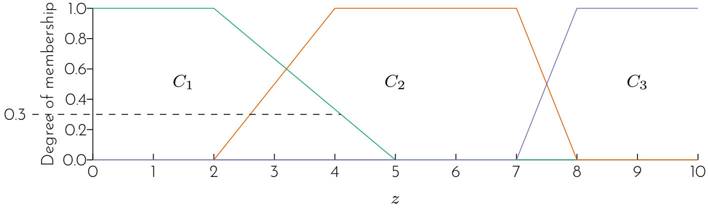

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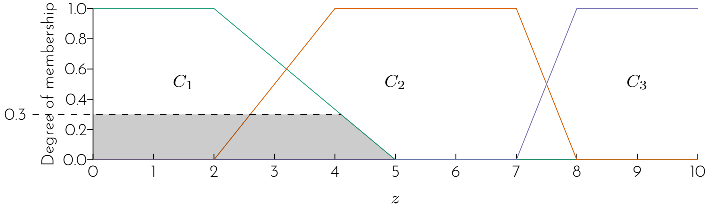

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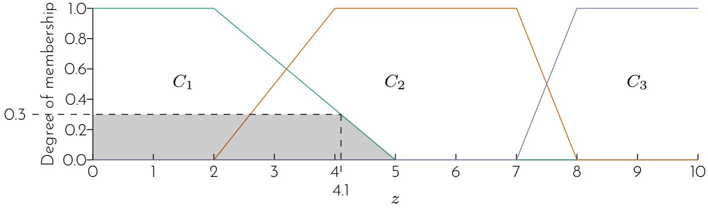
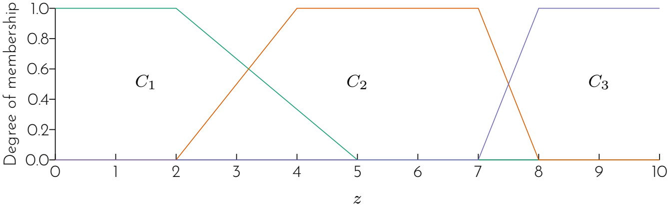


Example 2

Step 2: Rule evaluation (continued)

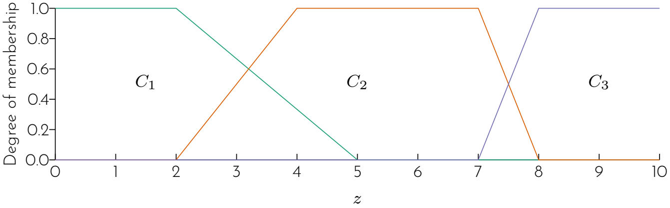
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is C_2	



Example 2

Step 2: Rule evaluation (continued)

Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is C_2	$\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

Rule 2		Degree of membership
IF	x is A_2	0.125
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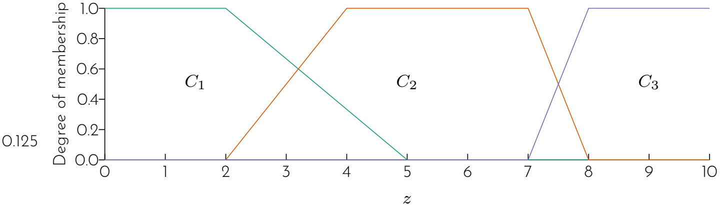
THEN	z is C_2	$\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

Rule 2		Degree of membership
IF	x is A_2	0.125
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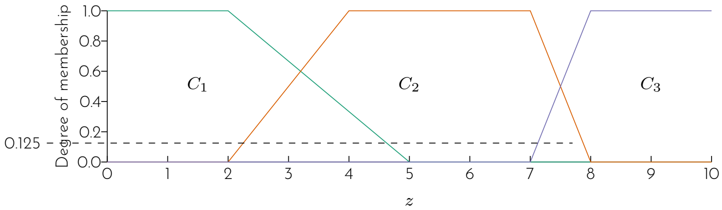

Example 2

Step 2: Rule evaluation (continued)

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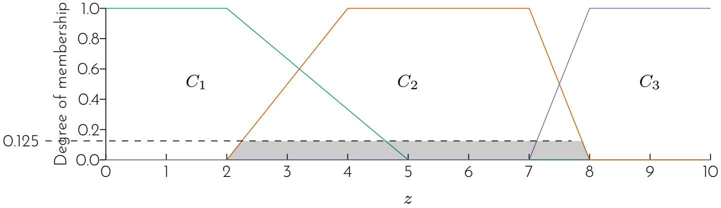

Example 2

Step 2: Rule evaluation (continued)

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IF	x is A_2	0.125
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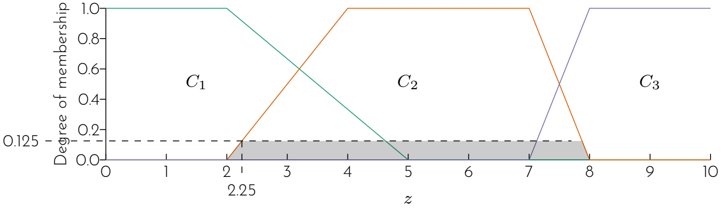
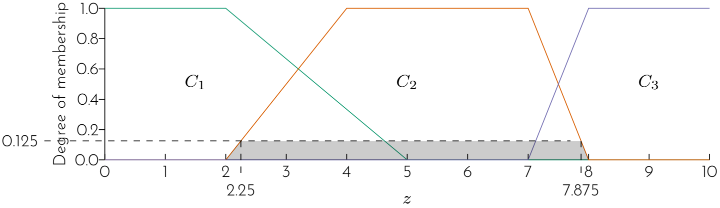


Example 2

Step 2: Rule evaluation (continued)

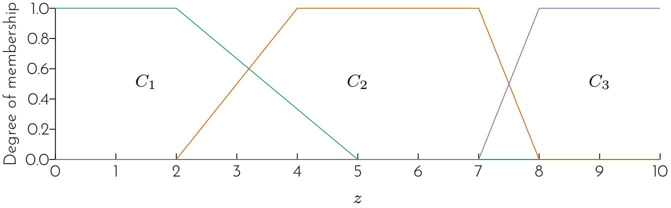
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is C_2	$\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

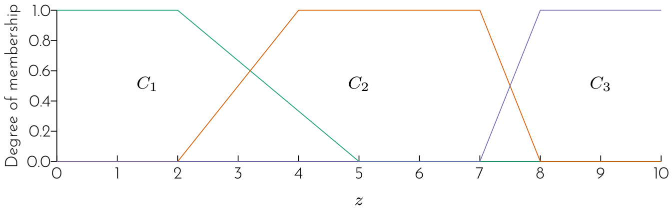
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	



Example 2

Step 2: Rule evaluation (continued)

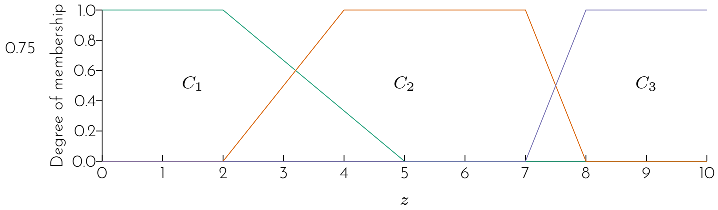
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	0.75



Example 2

Step 2: Rule evaluation (continued)

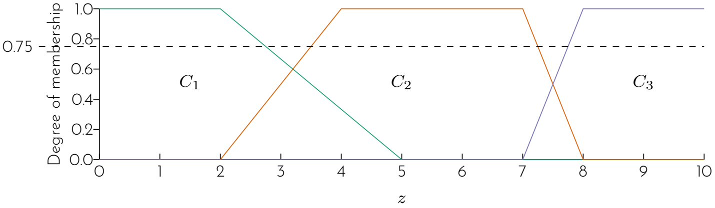
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	0.75



Example 2

Step 2: Rule evaluation (continued)

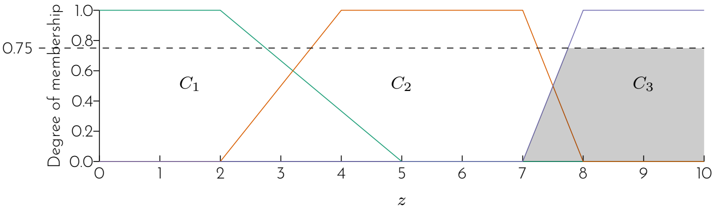
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	0.75



Example 2

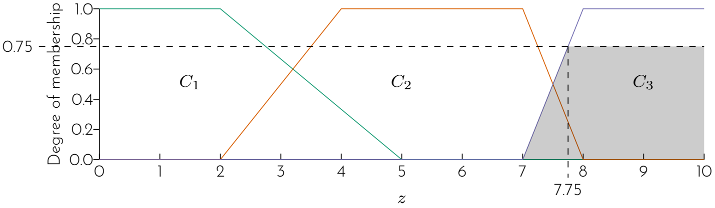
Step 2: Rule evaluation (continued)

Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	0.75



Step 2: Rule evaluation (continued)

Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is C_3	0.75



Example 2

Step 3: Aggregation of rule consequents

Example 2

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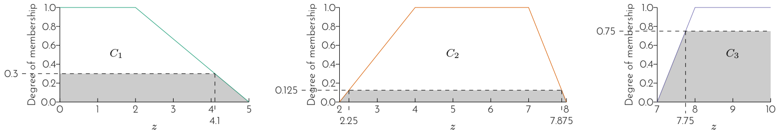
The clipped sets of the output variables are combined to form a single fuzzy set.

Example 2

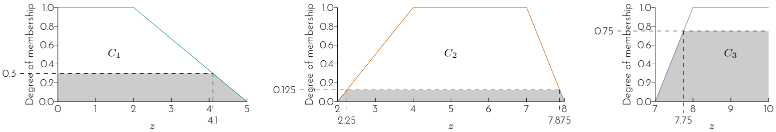
Step 3: Aggregation of rule consequents

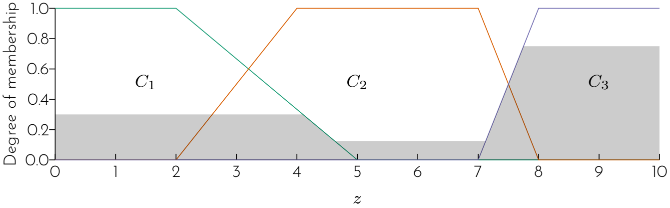
The clipped sets of the output variables are combined to form a single fuzzy set.



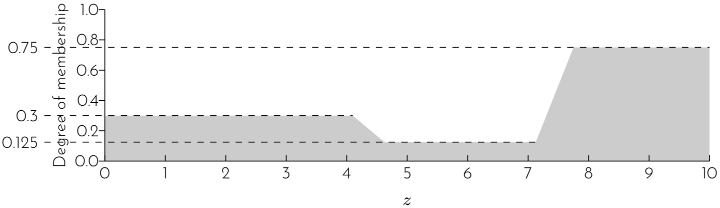
Step 3: Aggregation of rule consequents

The clipped sets of the output variables are combined to form a single fuzzy set.

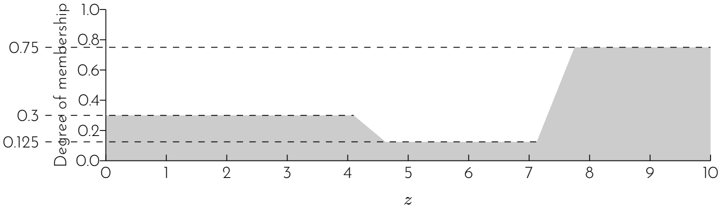




Example 2

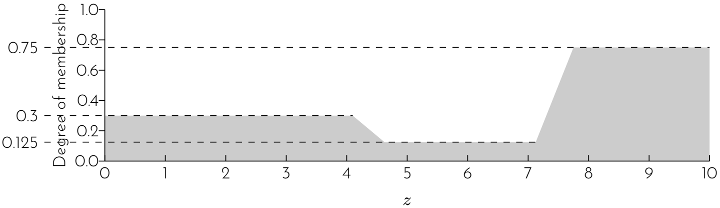

Degree of membership
1.0
0.8
0.75
0.6
0.4
0.3
0.2
0.125
0.0
0
1
2
3
4
5
6
7
8
9
10
z

Example 2



Step 4: Defuzzification

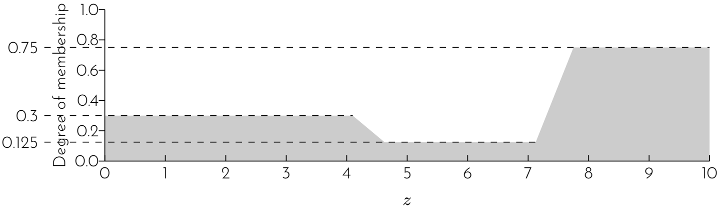
Example 2



Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

Example 2

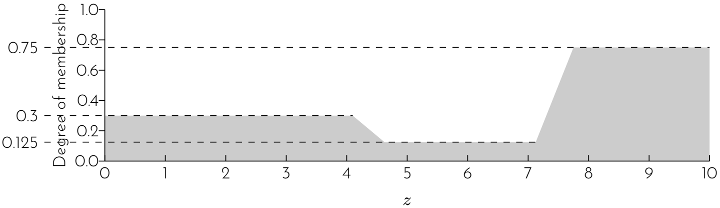


Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

$$\text{output} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A}$$

Example 2



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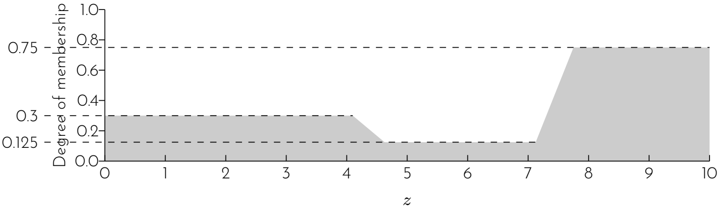
Defuzzification for Mamdani-style inference uses the centroid technique:

$$\text{output} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(0+1+2+3+4)(0.3)+(5+6+7)(0.125)+(8+9+10)(0.75)}{5(0.3)+3(0.125)+3(0.75)}$$

Example 2



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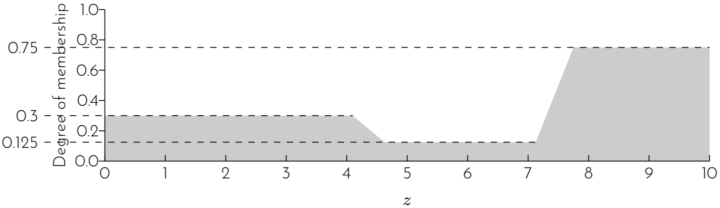
$$= 6.182$$

Example 2



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$$\text{output} = \frac{\sum_{x=a}^{b} \mu_A \cdot x}{\sum_{x=a}^{b} \mu_A} = \frac{(0+1+2+3+4)(0.3)+(5+6+7)(0.125)+(8+9+10)(0.75)}{5(0.3)+3(0.125)+3(0.75)}$$

$$= 6.182$$

Therefore, for $x_1 = 3.5$ and $y_1 = 6.5$, $z_1 = 6.182$

Example 3

Consider the fuzzy system with two input variables x and y, and one output variable z. All the variables have the range between 0 and 10.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$

Example 3

The following Sugeno-style rules apply to this system.

Rule 1

IF x is A_3
OR y is B_1
THEN z is 3

Rule 2

IF x is A_2
AND y is B_2
THEN z is 5

Rule 3

IF x is A_1
THEN z is 8

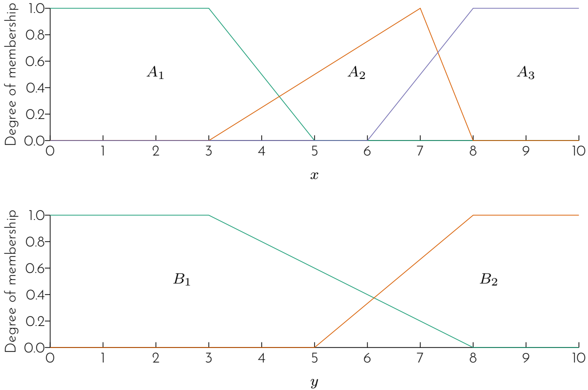
Given that x_1 is 3.5, y_1 is 6.5, what is the value of z_1?

Example 3

First, draw the membership function of the fuzzy sets on the graph paper.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$

Example 3

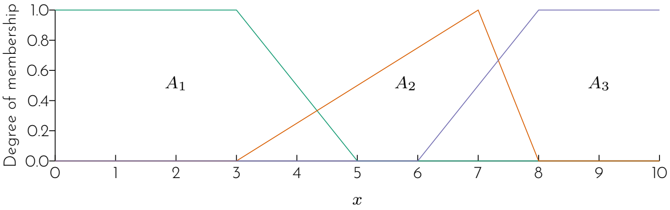


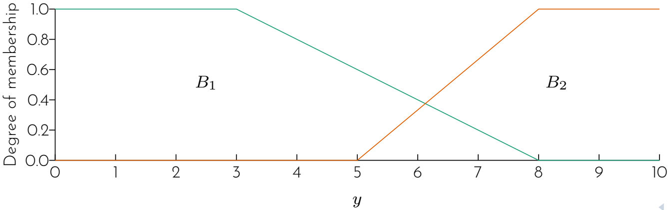
Degree of membership
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9 10
x
A_1
A_2
A_3
Degree of membership
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9 10
y
B_1
B_2

Example 3

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.

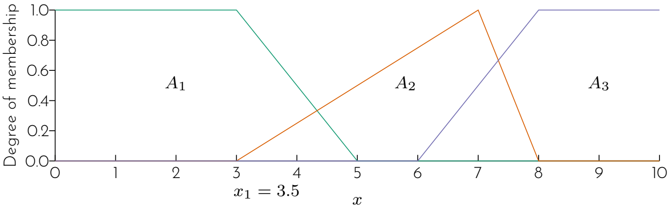


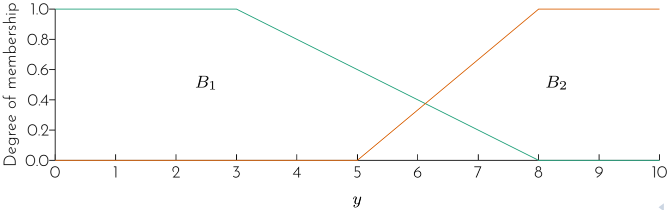


Example 3

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.





Example 3

Step 1: Fuzzification

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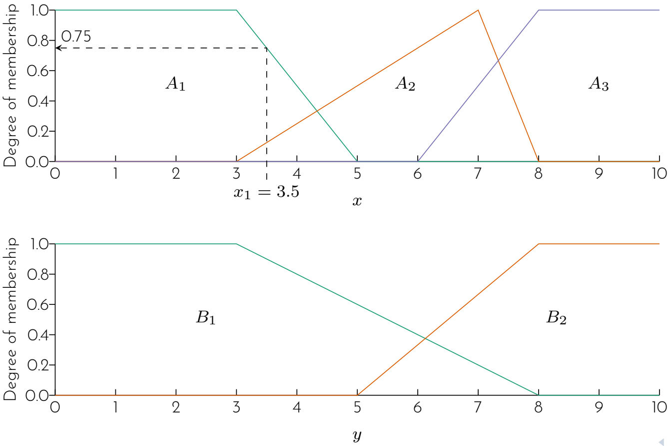

Step 1: Fuzzification

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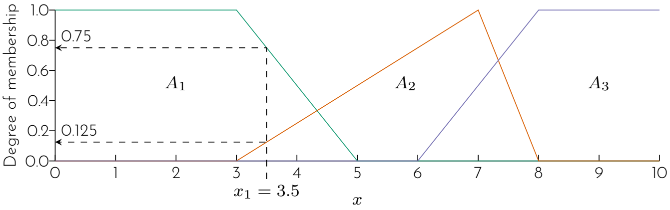
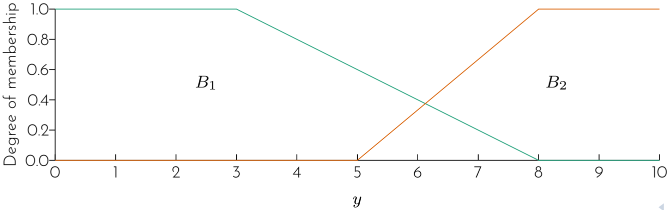
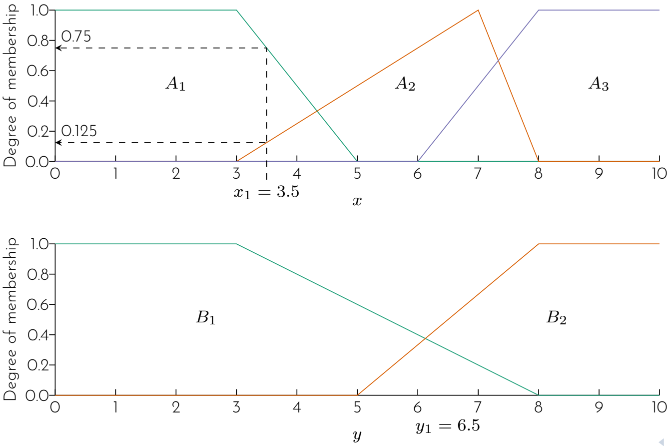




Example 3

Step 1: Fuzzification

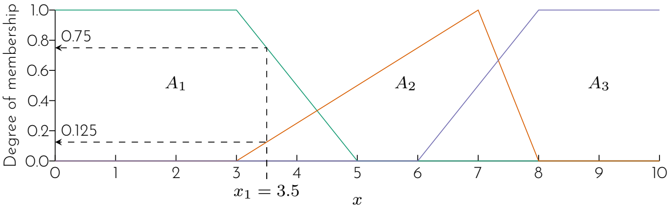
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.

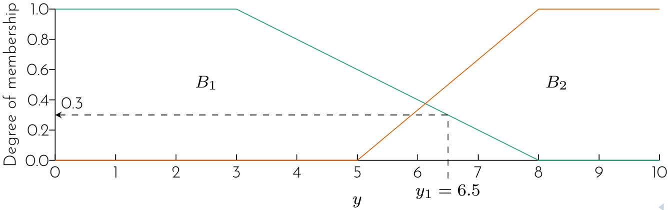


Example 3

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.

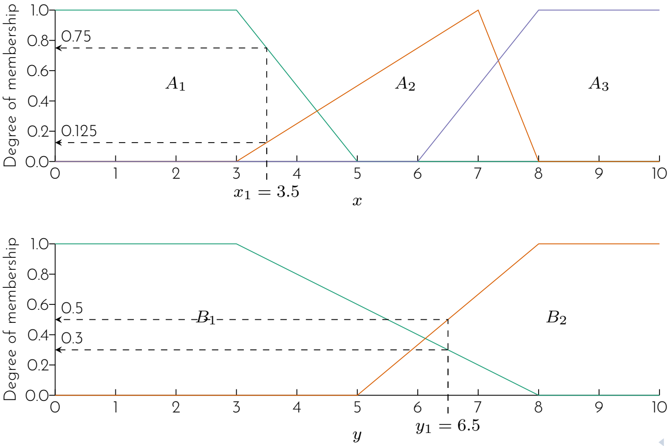




Example 3

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification (continued)

Therefore we have

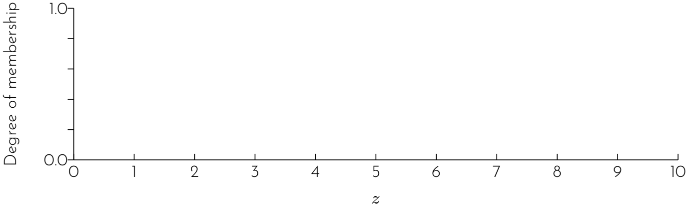
$$\mu_{x=A_1} = 0.75$$
$$\mu_{x=A_2} = 0.125$$
$$\mu_{x=A_3} = 0$$

$$\mu_{y=B_1} = 0.3$$
$$\mu_{y=B_2} = 0.5$$

Example 3

Step 2: Rule evaluation

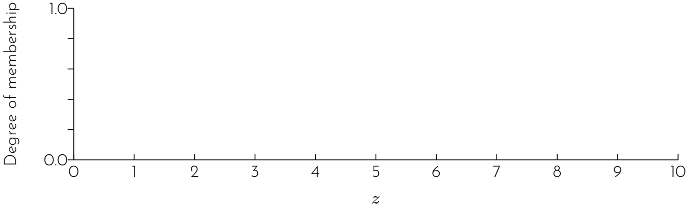
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	



Example 3

Step 2: Rule evaluation

Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	$\max(0.0, 0.3) = 0.3$



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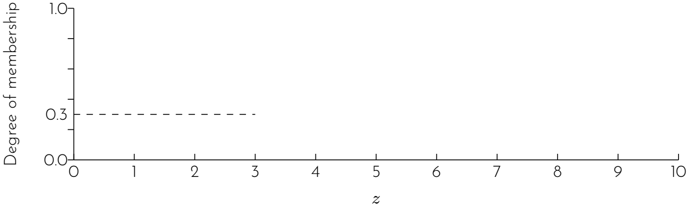

Example 3

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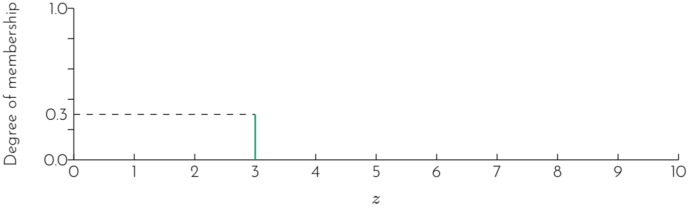
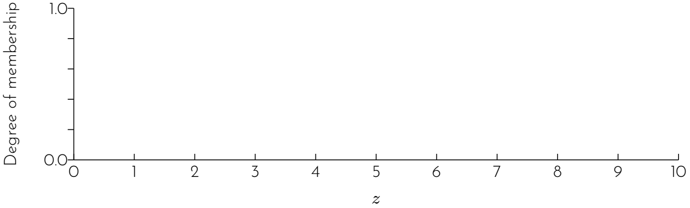


Example 3

Step 2: Rule evaluation (continued)

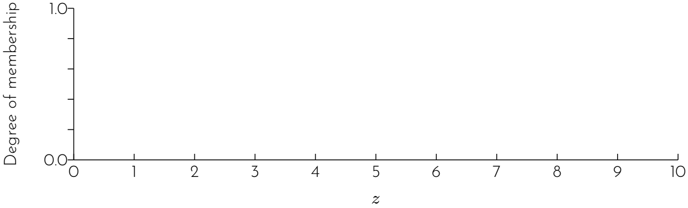
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	



Example 3

Step 2: Rule evaluation (continued)

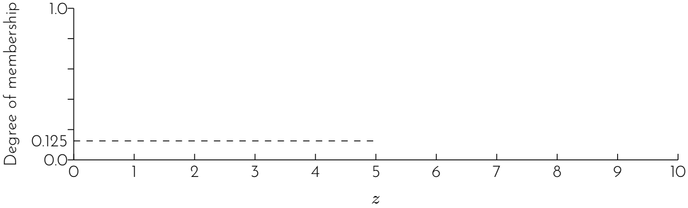
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Example 3

Step 2: Rule evaluation (continued)

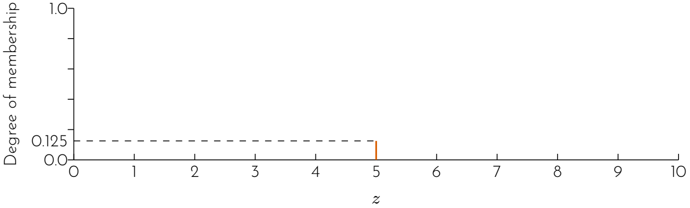
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Example 3

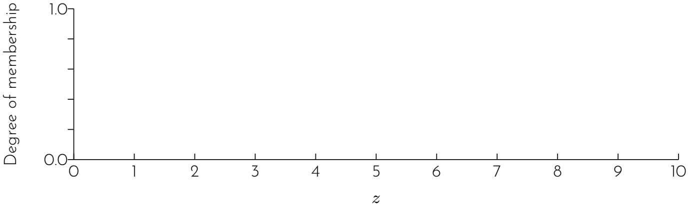
Step 2: Rule evaluation (continued)

Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Step 2: Rule evaluation (continued)

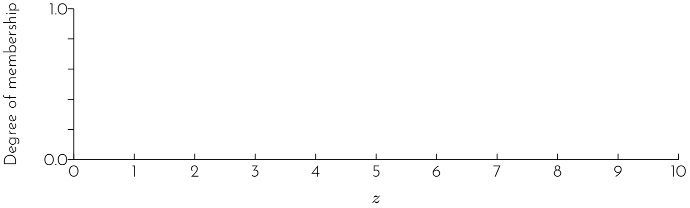
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	



Example 3

Step 2: Rule evaluation (continued)

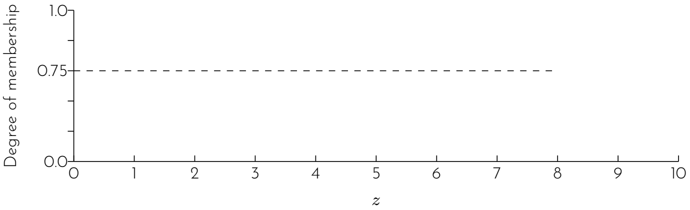
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Example 3

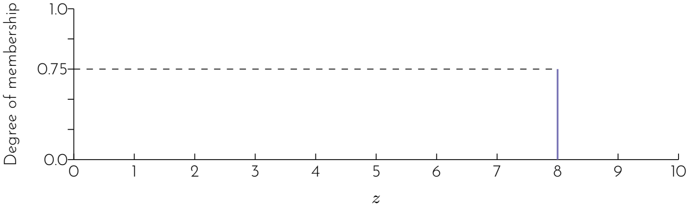
Step 2: Rule evaluation (continued)

Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Step 2: Rule evaluation (continued)

Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Example 3

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The outputs of all fuzzy rules are combined.

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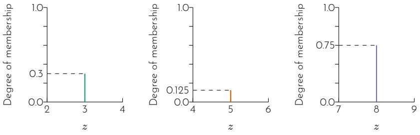

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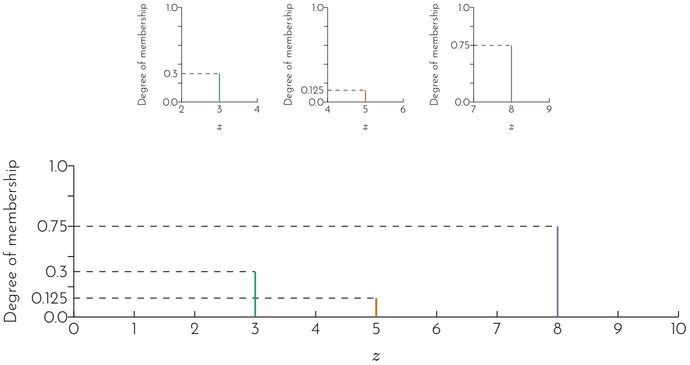
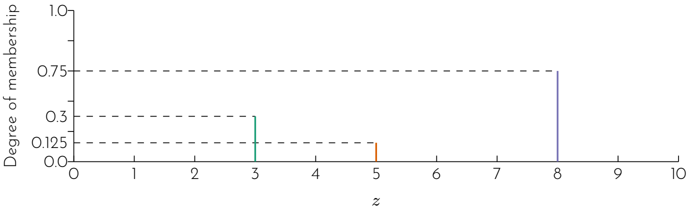
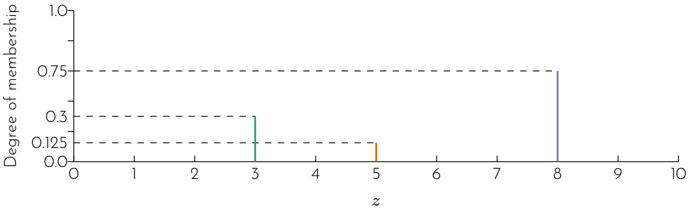


Example 3

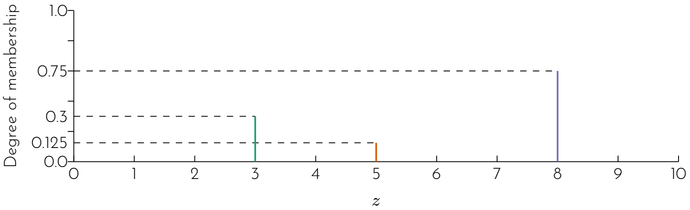


Example 3



Step 4: Defuzzification

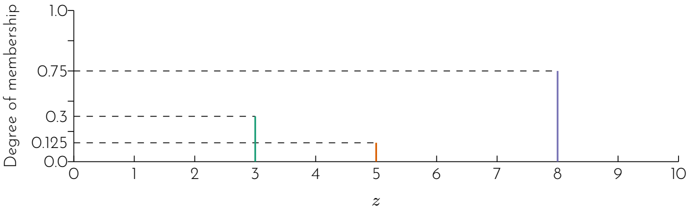
Example 3



Step 4: Defuzzification

Defuzzification for Sugeno-style inference uses the weighted average technique:

Example 3



Step 4: Defuzzification

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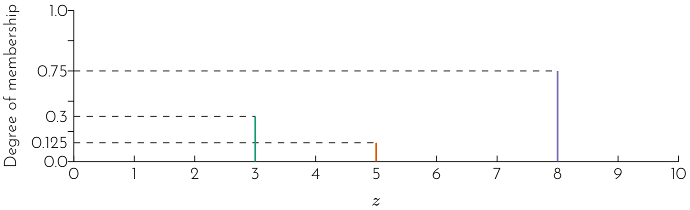
$$\text{output} = \frac{\sum_{x=a}^{b} \alpha_x \cdot x}{\sum_{x=a}^{b} \alpha_x}$$

Example 3



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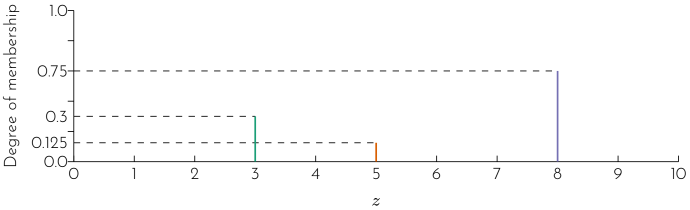
$$\text{output} = \frac{\sum_{x=a}^{b} \alpha_x \cdot x}{\sum_{x=a}^{b} \alpha_x} = \frac{(3)(0.3) + (5)(0.125) + (8)(0.75)}{0.3 + 0.125 + 0.75}$$

Example 3



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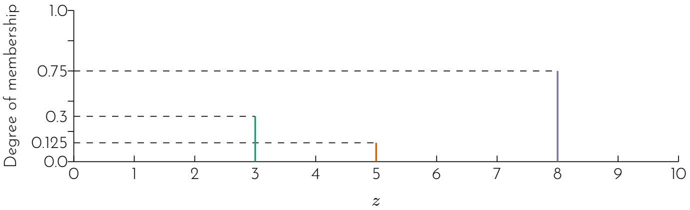
$$= 6.404$$

Example 3



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$$= 6.404$$

Therefore, for $x_1 = 3.5$ and $y_1 = 6.5$, $z_1 = 6.404$

What are the variables that can be represented as fuzzy variables?

Building a fuzzy expert system

The typical process in developing a fuzzy expert system incorporates the following steps:

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The typical process in developing a fuzzy expert system incorporates the following steps:

1. Specify the problem and define linguistic variables.
2. Determine fuzzy sets.
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4. Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.
5. Evaluate and tune the system.

Case study 4

Build a fuzzy expert system to control the speed of a train.

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Build a fuzzy expert system to control the speed of a train.

Step 1: Specify the problem and define linguistic variables.

Objective:

Case study 4

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Design system to smoothly slow and stop train from any speed and distance from station.

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Linguistic variables:

Case study 4

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Linguistic variables:

Input

Case study 4

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Linguistic variables:

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Output

Case study 4

Build a fuzzy expert system to control the speed of a train.

Step 1: Specify the problem and define linguistic variables.

Objective:

Design system to smoothly slow and stop train from any speed and distance from station.

Linguistic variables:

Input – speed, distance

Output – brake, throttle

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed

Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed

Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance

Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed

Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance

Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Throttle

Linguistic value	Numerical range
Full	60% - 100%
Medium	20% - 80%
Slight	3% - 30%
Very slight	1% - 5%
No	0% - 2%

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed

Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance

Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Throttle

Linguistic value	Numerical range
Full	60% - 100%
Medium	20% - 80%
Slight	3% - 30%
Very slight	1% - 5%
No	0% - 2%

Brake

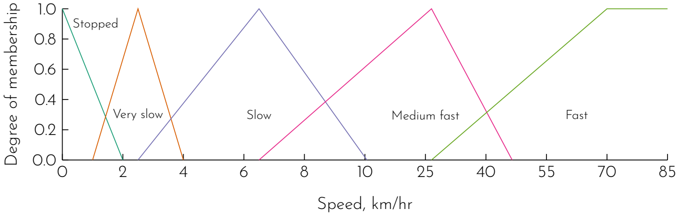
Linguistic value	Numerical range
Full	98% - 100%
Medium	95% - 99%
Slight	70% - 97%
Very slight	20% - 80%
No	0% - 40%

Case study 4

Step 2: Determine fuzzy sets.

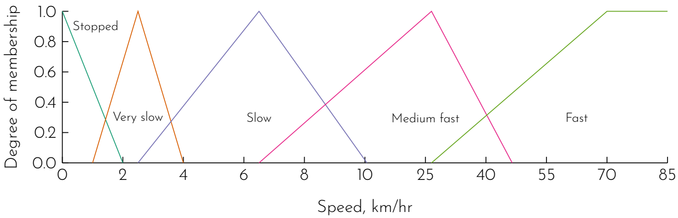
Case study 4

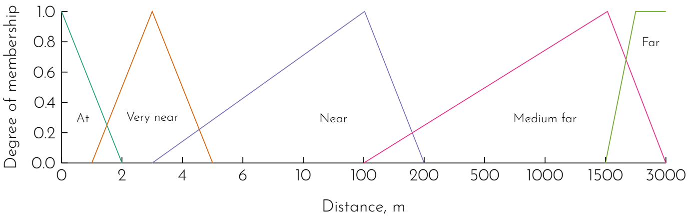
Step 2: Determine fuzzy sets.



Case study 4

Step 2: Determine fuzzy sets.



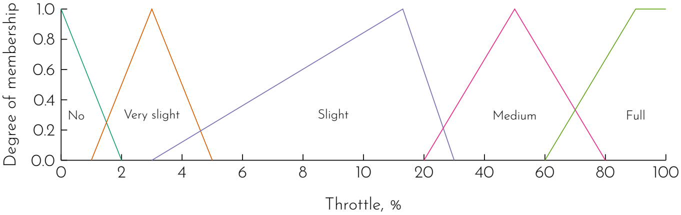


Case study 4

Step 2 (continued)

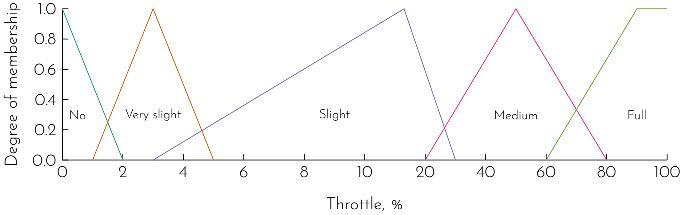
Case study 4

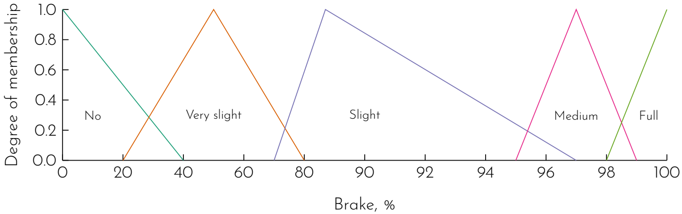
Step 2 (continued)



Case study 4

Step 2 (continued)





Step 3: Elicit and construct fuzzy rules.

Case study 4

Step 3: Elicit and construct fuzzy rules.

Fuzzy association memory (FAM) representation

Case study 4

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Fuzzy association memory (FAM) representation

- Represent fuzzy rules in matrix form.

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Fuzzy association memory (FAM) representation

- Represent fuzzy rules in matrix form.
- Each dimension of the matrix represents an input variable.
- The number of row/column along each dimension equals to the number of subsets in the corresponding input variable.

Step 3: Elicit and construct fuzzy rules.

Fuzzy association memory (FAM) representation

- Represent fuzzy rules in matrix form.
- Each dimension of the matrix represents an input variable.
- The number of row/column along each dimension equals to the number of subsets in the corresponding input variable.
- The elements in each cell is the rule consequent.

Case study 4

Step 3 (continued)

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Each cell provides the rule consequent for the intersection (AND) of the corresponding input values.

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Each cell provides the rule consequent for the intersection (AND) of the corresponding input values.

For example, the top left cell represents the rule consequent of 'speed' is 'stopped' AND 'distance' is 'at'.

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped	Full brake No throttle	Full brake VS throttle			
	Very slow	Full brake No throttle	Medium brake VS throttle	Slight brake VS throttle		
	Slow	Full brake No throttle	Medium brake VS throttle	VS brake Slight throttle		
	Medium fast				VS brake Med. throttle	No brake Full throttle
	Fast				VS brake Med. throttle	No brake Full throttle

Case study 4

Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

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- This step concerns of actually building the expert system using a library or toolbox.

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- This step concerns of actually building the expert system using a library or toolbox.
- It also includes making decision on how the rules are evaluated. In most cases, we will be using the centroid method.

Case study 4

Step 4 (continued)

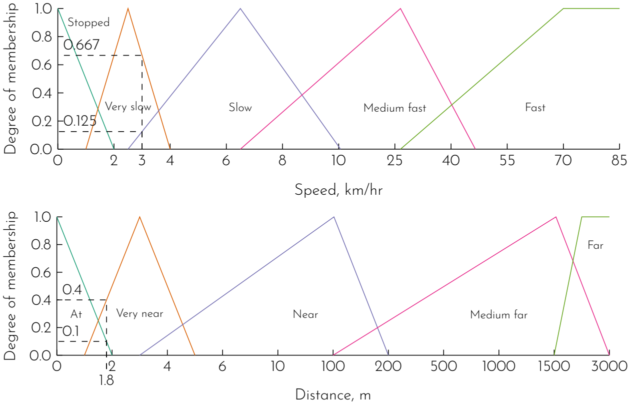
To illustrate how the rules are evaluated, we consider the inputs of

$$\text{speed} = 3 \text{ km/hr}$$
$$\text{distance} = 1.8 \text{ m}$$

Case study 4

Step 4 (continued)

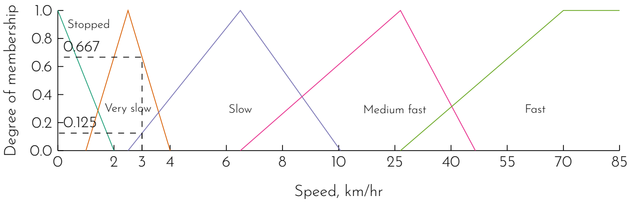
With speed = 3 km/hr and distance = 1.8 m,

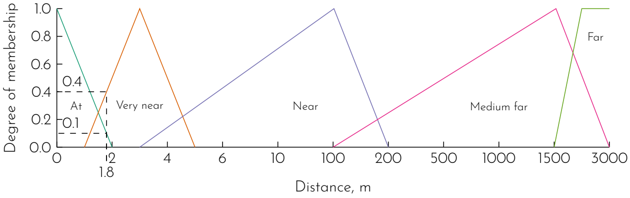


Case study 4

Step 4 (continued)

With speed = 3 km/hr and distance = 1.8 m,





From the fuzzy set diagrams, we have

$$\mu_{\text{speed=stopped}} = 0$$

$$\mu_{\text{speed=very slow}} = 0.667$$

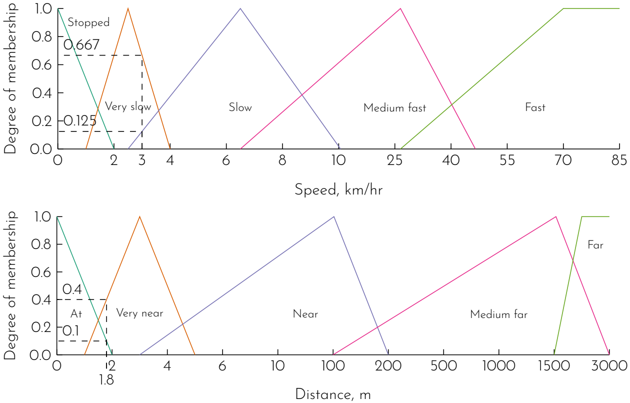
$$\mu_{\text{speed=slow}} = 0.125$$

$$\mu_{\text{speed=medium fast}} = 0$$

$$\mu_{\text{speed=fast}} = 0$$

Step 4 (continued)

With speed = 3 km/hr and distance = 1.8 m,



From the fuzzy set diagrams, we have

$$\mu_{\text{speed=stopped}} = 0$$
$$\mu_{\text{speed=very slow}} = 0.667$$
$$\mu_{\text{speed=slow}} = 0.125$$
$$\mu_{\text{speed=medium fast}} = 0$$
$$\mu_{\text{speed=fast}} = 0$$

$$\mu_{\text{distance=at}} = 0.1$$
$$\mu_{\text{distance=very near}} = 0.4$$
$$\mu_{\text{distance=near}} = 0$$
$$\mu_{\text{distance=medium far}} = 0$$
$$\mu_{\text{distance=far}} = 0$$

Case study 4

Step 4 (continued)

Case study 4

Step 4 (continued)

Since we only have non-zero degree of membership for

speed = very slow

speed = slow

distance = at

distance = very near

Case study 4

Step 4 (continued)

Since we only have non-zero degree of membership for

speed = very slow

speed = slow

distance = at

distance = very near

Only the shaded rules are fired.

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped	Full brake No throttle	Full brake VS throttle			
	Very slow	Full brake No throttle	Medium brake VS throttle	Slow brake VS throttle		
	Slow	Full brake No throttle	Medium brake VS throttle	VS brake Slow throttle		
	Medium fast				VS brake Med. throttle	No brake Full throttle
	Fast				VS brake Med. throttle	No brake Full throttle

Case study 4

Step 4 (continued)

Case study 4

Step 4 (continued)

The four rules that fired are

Case study 4

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Rule 3

IF	speed = slow	0.125
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Rule 3

IF	speed = slow	0.125
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

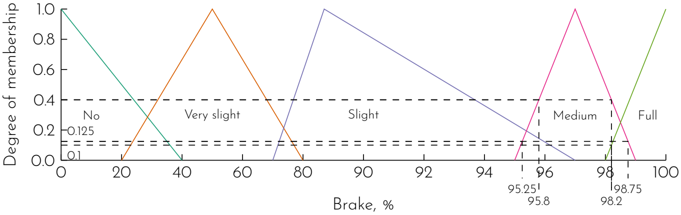
Rule 4

IF	speed = slow	0.125
AND	distance = very near	0.4
THEN	brake = medium	0.125
	throttle = very slight	0.125

Case study 4

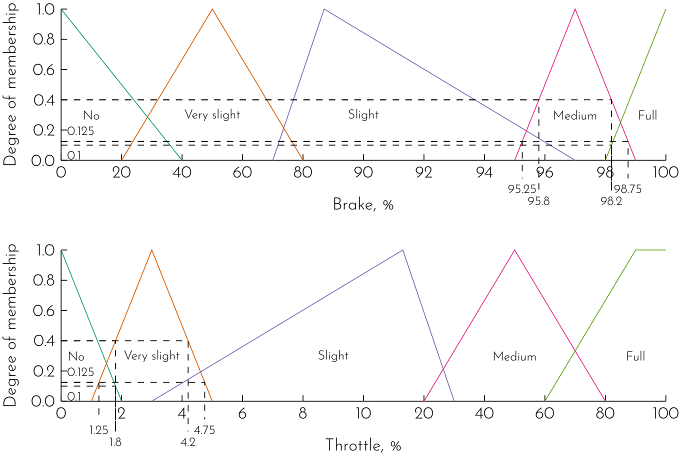
Step 4 (continued)

Step 4 (continued)



Case study 4

Step 4 (continued)



Case study 4

Step 4 (continued)

Case study 4

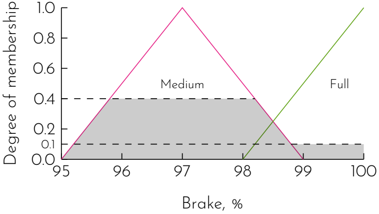
Step 4 (continued)

Aggregation of outputs

Case study 4

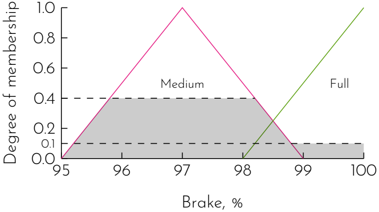
Step 4 (continued)

Aggregation of outputs



Step 4 (continued)

Aggregation of outputs



brake

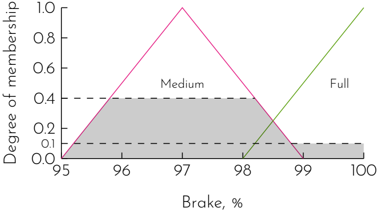
$$= \frac{(95)(0) + (96+97+98)(0.4) + (99+100)(0.1)}{0+3(0.4)+2(0.1)}$$

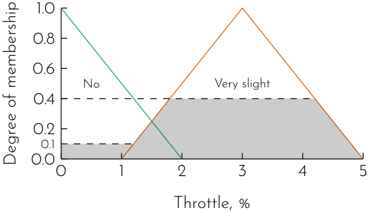
$$= 97.36\%$$

Case study 4

Step 4 (continued)

Aggregation of outputs





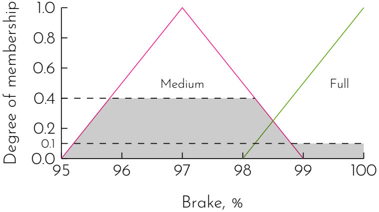
brake

$$= \frac{(95)(0) + (96+97+98)(0.4) + (99+100)(0.1)}{0 + 3(0.4) + 2(0.1)}$$

$$= 97.36\%$$

Step 4 (continued)

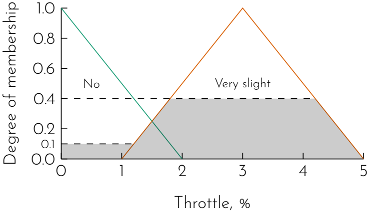
Aggregation of outputs



brake

$$= \frac{(95)(0) + (96+97+98)(0.4) + (99+100)(0.1)}{0 + 3(0.4) + 2(0.1)}$$

$$= 97.36\%$$



throttle

$$= \frac{(0+1)(0.1) + (2+3+4)(0.4) + (5)(0)}{2(0.1) + 3(0.4) + 0}$$

$$= 2.64\%$$

Case study 4

Step 5: Evaluate and tune the system

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- This step is to identify if the designed fuzzy system meets the requirements of the specified problem.

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- Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.

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- Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.
- Another method of evaluation is to generate surface plots of the rules. This can be achieved using appropriate tools.

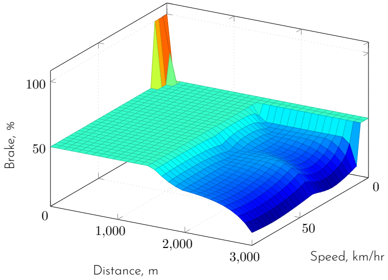
Step 5: Evaluate and tune the system

- This step is to identify if the designed fuzzy system meets the requirements of the specified problem.
- Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.
- Another method of evaluation is to generate surface plots of the rules. This can be achieved using appropriate tools.
- What parameters of the system are we tuning?

Case study 4

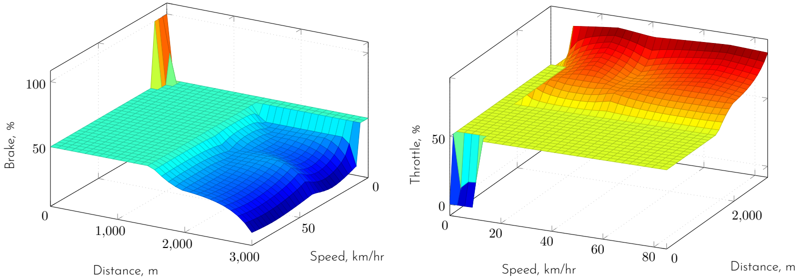
Step 5 (continued)

Step 5 (continued)



Case study 4

Step 5 (continued)



Case study 5

Build a fuzzy expert system to identify the percentage of tips a customer will give based on the service and the food the customer received.

Case study 5

Step 1: Specify the problem and define linguistic variables

Case study 5

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Case study 5

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

- Input

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input – service, food

Case study 5

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input – service, food

Output

Case study 5

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input – service, food

Output – tips

Case study 5

Step 1 (continued)

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Service

Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Service

Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Food

Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Step 1 (continued)

Define the range of the linguistic variables.

Service

Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Food

Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

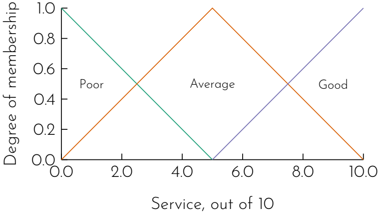
Tips

Linguistic value	Numerical range
Low	0% - 15%
Medium	0% - 30%
High	15% - 30%

Case study 5

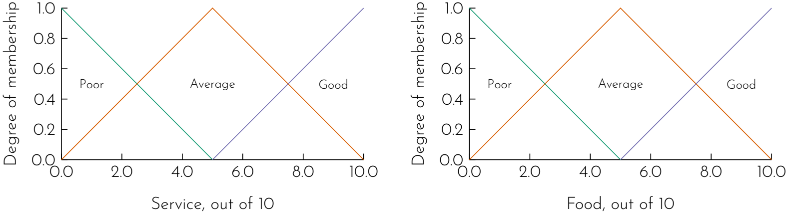
Step 2: Determine fuzzy sets.

Step 2: Determine fuzzy sets.



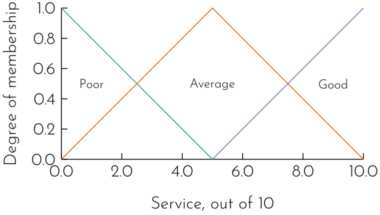
Case study 5

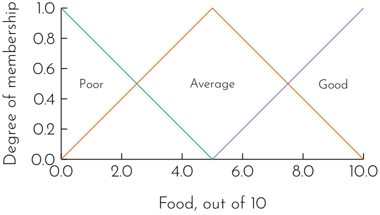
Step 2: Determine fuzzy sets.

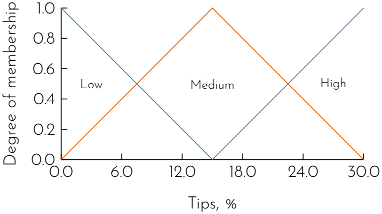


Case study 5

Step 2: Determine fuzzy sets.







Case study 5

Step 3: Elicit and construct fuzzy rules.

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Step 3: Elicit and construct fuzzy rules.

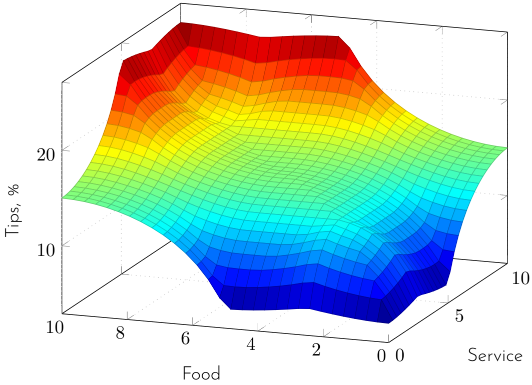
		Food		
		Poor	Average	Good
Service	Poor	low tips	low tips	medium tips
	Average	low tips	medium tips	high tips
	Good	medium tips	high tips	high tips

Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

We can use Matlab Fuzzy Logic Toolbox to encode our fuzzy system.

Step 5: Evaluate and tune the system

The outputs of the fuzzy system are plotted against the range of the inputs (0-10).



Case study 5

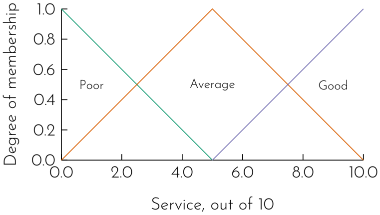
Step 5 (continued)

Step 5 (continued)

If I'm not happy with the result, I might modify the degree of membership in the score of service such that

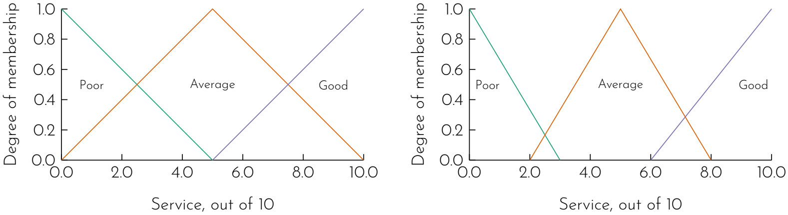
Step 5 (continued)

If I'm not happy with the result, I might modify the degree of membership in the score of service such that

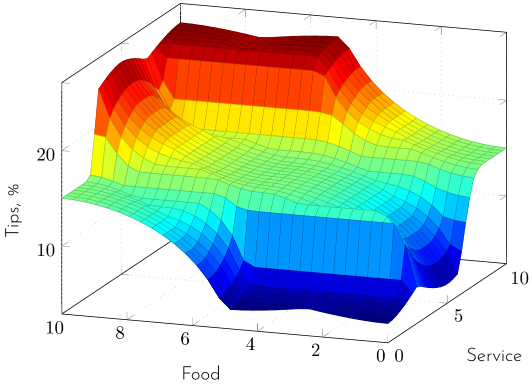


Step 5 (continued)

If I'm not happy with the result, I might modify the degree of membership in the score of service such that



Step 5 (continued)



Case study 6

Consider a problem of operating a service centre of spare parts.

A service centre keeps spare parts and repairs failed ones. A customer brings a failed item and receives a spare of the same type. Failed parts are repaired, placed on the shelf, and thus become spares. If the required spare is available on the shelf, the customer takes it and leaves the service centre. However, if there is no spare on the shelf, the customer has to wait until the needed item becomes available. The objective here is to advise a manager of the service centre on certain decision policies to keep the customers satisfied.

Case study 6

Step 1: Specify the problem and define linguistic variables

Case study 6

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

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Case study 6

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Objective :

Case study 6

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Objective : Advise the manager of the service centre of the number of spares required to maintain the mean delay in customer service within an acceptable range.

Case study 6

Step 1 (continued)

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Case study 6

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Mean delay (normalised), m

Linguistic value	Numerical range
very short	0 - 0.3
short	0.1 - 0.5
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Case study 6

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Mean delay (normalised), m

Linguistic value	Numerical range
very short	0 - 0.3
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Number of servers (normalised), s

Linguistic value	Numerical range
small	0 - 0.35
medium	0.3 - 0.7
large	0.6 - 1

Step 1 (continued)

Define the range of the linguistic variables.

Mean delay (normalised), m

Linguistic value	Numerical range
very short	0 - 0.3
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Number of servers (normalised), s

Linguistic value	Numerical range
small	0 - 0.35
medium	0.3 - 0.7
large	0.6 - 1

Utilisation factor, ρ

Linguistic value	Numerical range
low	0 - 0.6
medium	0.4 - 0.8
high	0.6 - 1

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Number of spares (normalised), n

Linguistic value	Numerical range
very small	0 - 0.30
small	0 - 0.40
rather small	0.25 - 0.45
medium	0.30 - 0.70
rather large	0.55 - 0.75
large	0.60 - 1
very large	0.70 - 1

Case study 6

Step 2: Determine fuzzy sets.

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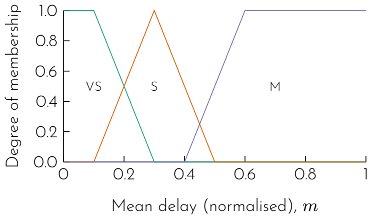

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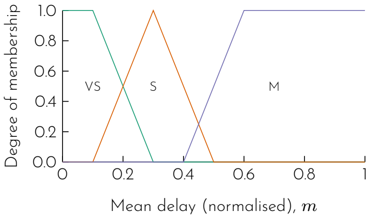

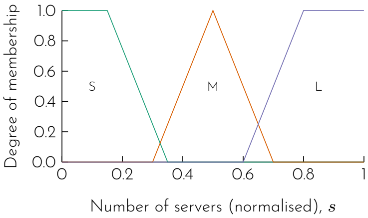

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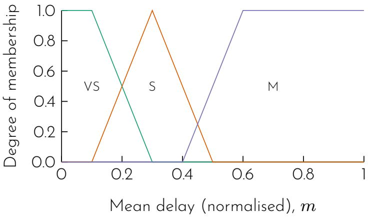

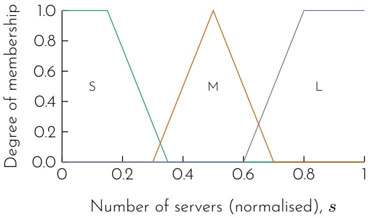

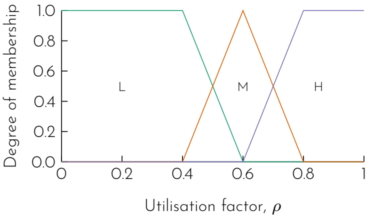

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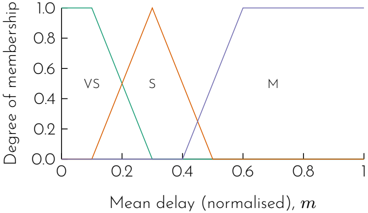

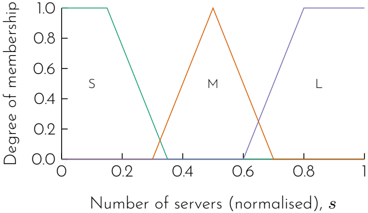

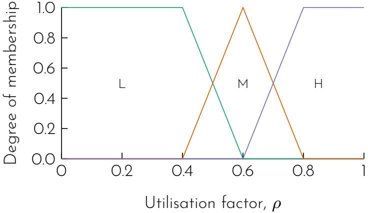

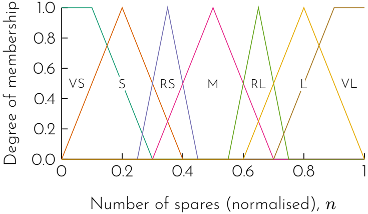

Case study 6

Step 3: Elicit and construct fuzzy rules.

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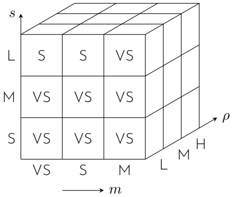

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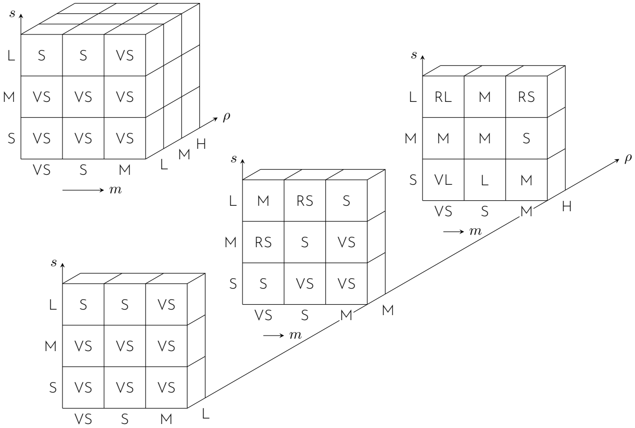

Case study 6

Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

Case study 6

Step 5: Evaluate and tune the system

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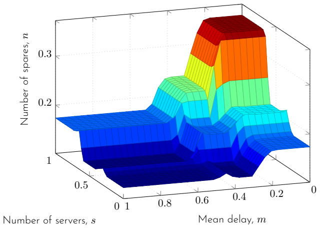
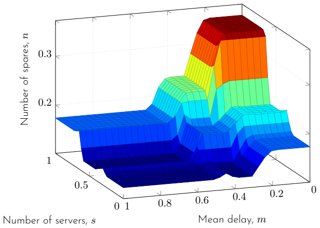
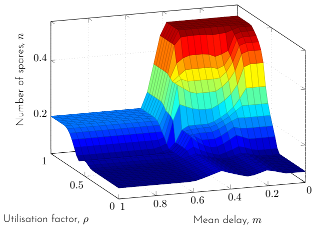


Case study 6

Step 5: Evaluate and tune the system





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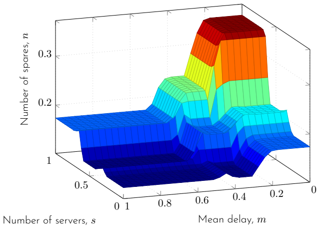

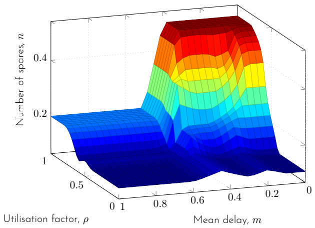

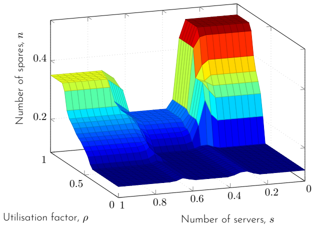

Case study 6

Step 5 (continued)

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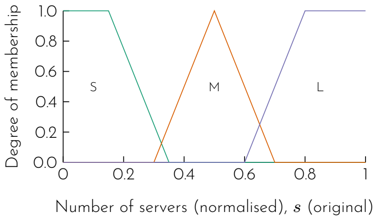

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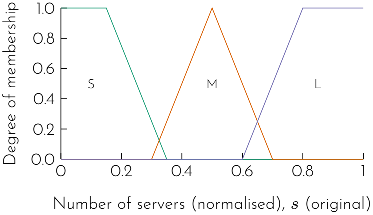

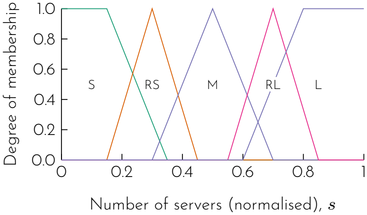

Case study 6

Step 5 (continued)

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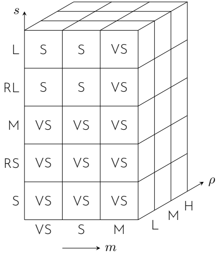

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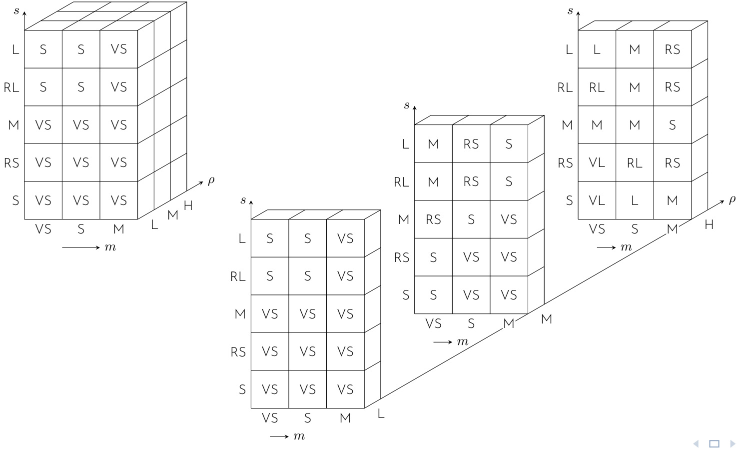

Case study 6

Step 5 (continued)

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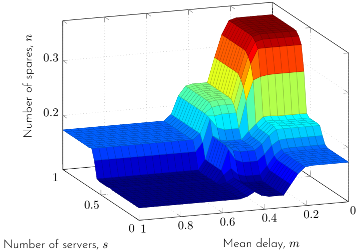
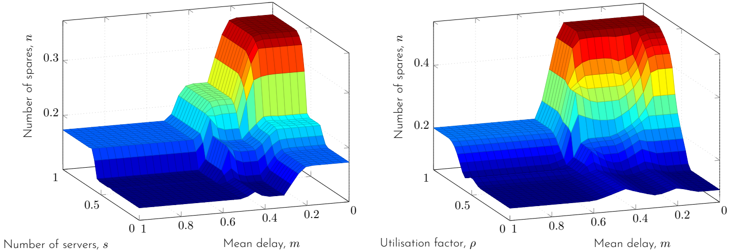
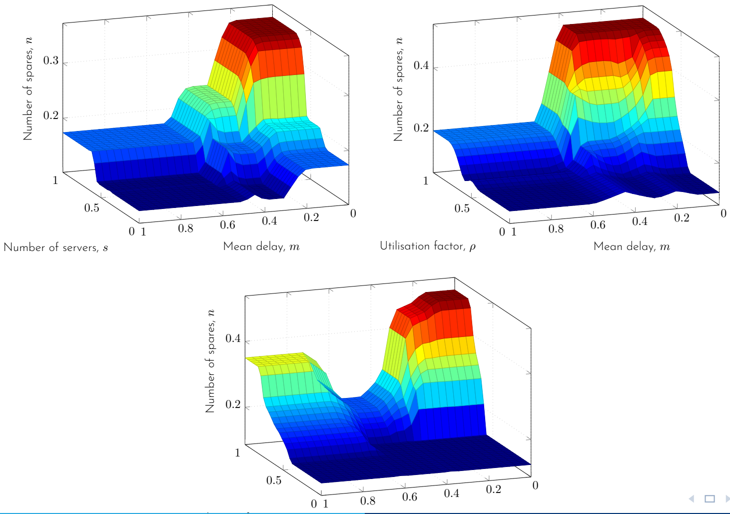


Step 5 (continued)



Case study 6

Step 5 (continued)



Example 7

Construct Sugeno-style fuzzy rules with the inputs of

- temperature of the room,
- number of people in the room,
- ambient temperature (temperature outside the room), and
- time to spend in the room,

and output of cooling speed of an air conditioner.

Perform a inference for temperature of 30°C, 2pm, 20 people in the room, ambient temperature of 33°C, and to spend 1 hour in the room.

Example 8

How would it be different if you were to construct a Mamdani-style fuzzy system?

CSC3034 Current Course Update

03 Fuzzy Inference and Fuzzy Systems - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 2 uses scikit-fuzzy to build and evaluate Mamdani control systems.
- The physical-AI extension applies fuzzy brake/throttle logic to a simulated robot.
- The updated solution checks optional dependencies and supports a standalone fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.